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Pregnancy and Kidney Disease

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CHAPTER OUTLINE

KEY POINTS

- Preeclampsia is now considered a major risk factor for long-term cardiovascular disease, including stroke, end-stage kidney disease, and cardiovascular death. Women with a history of preeclampsia should be screened and treated for modifiable risk factors, such as diabetes, hypertension, tobacco use, and obesity.
- The 2013 American College of Obstetrics and Gynecology diagnostic criteria for preeclampsia include hypertension and EITHER proteinuria OR another feature of severe preeclampsia (Table 48.3).
- Intensive dialysis (≥36 hours/week) is associated with improved fertility and pregnancy outcomes in women with end-stage renal disease.
- Medications that are safe in pregnant transplant recipients include calcineurin inhibitors (tacrolimus and cyclosporine), azathioprine, and prednisone. Rapamycin and mycophenolate are teratogenic and contraindicated in pregnancy.

INTRODUCTION

Pregnancy is characterized by a myriad of physiologic changes, of which the emergence of a placenta and growing fetus are the most dramatic. Hypertension and renal disease occurring in the setting of pregnancy present a unique set of clinical challenges. This chapter will include a detailed discussion of preeclampsia, a syndrome specific to pregnancy that remains one of the most enigmatic human disorders and continues to claim the lives of thousands of mothers and neonates yearly. The approach to acute renal failure in pregnancy will also be discussed. Current data on epidemiology and management of chronic hypertension, chronic renal disease, glomerular disease, and kidney transplantation in pregnancy will be reviewed. This chapter will offer the reader insights into the emerging understanding of the pathogenesis of preeclampsia, and provide a sound basis for the management of pregnancy from a nephrologist's perspective.

PHYSIOLOGIC CHANGES OF PREGNANCY

HEMODYNAMIC AND VASCULAR CHANGES OF NORMAL PREGNANCY

Normal pregnancy is characterized by profound vascular and hemodynamic changes that reach far beyond the fetus and placenta (Table 48.1). Early in pregnancy, systemic vascular resistance (SVR) decreases and arterial compliance increases.[1] These changes are evident by 6 weeks' gestation, prior to the establishment of the uteroplacental circulation.[2] This decrease in SVR leads directly to several other cardiovascular changes. Mean arterial blood pressure (BP) falls by an average of 10 mm Hg, reaching its lowest point at 18–24 weeks' gestation (Fig. 48.1). Sympathetic activity is increased, reflected in a 15% to 20% increase in heart rate.[3] The combination of increased heart rate and decreased afterload leads to a large increase in cardiac output in the early first trimester, which peaks at 50% above prepregnancy levels by the middle of the third trimester (Fig. 48.2).

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Table 48.1 Physiologic Changes in Pregnancy

Physiologic Variable	Change in Pregnancy
Hemodynamic Parameters	
Plasma volume	Increases 30%–50% above baseline
Blood pressure	Decreases by approximately 10 mm Hg below prepregnancy level, with nadir in second trimester; gradual increase toward prepregnancy levels by term
Cardiac output	Increases 30%–50%
Heart rate	Increases by 15–20 beats/min
Renal blood flow	Increases to 80% above baseline
Glomerular filtration rate	150–200 mL/min (increases to 40%–50% above baseline)
Serum Chemistry and Hematologic Changes	
Hemoglobin	Decreases by an average of 2 g/L (from 13 to 11 g/L) owing to plasma volume expansion out of proportion to the increase in red blood cell mass
Creatinine	Decreases to 0.4–0.5 mg/dL
Uric acid	Decreases to a nadir of 2.0–3.0 mg/dL by 22–24 weeks, then increases back to nonpregnant levels toward term
pH	Increases slightly to 7.44
Partial pressure of carbon dioxide (pCO_2)	Decreases by approximately 10 mm Hg to an average of 27–32 mm Hg
Calcium	Increased calcitriol stimulates increases in both intestinal calcium reabsorption and urinary calcium excretion
Sodium	Decreases by 4–5 mEq/L below nonpregnancy levels
Osmolality	Decreases to a new osmotic set point of approximately 270 mOsm/kg

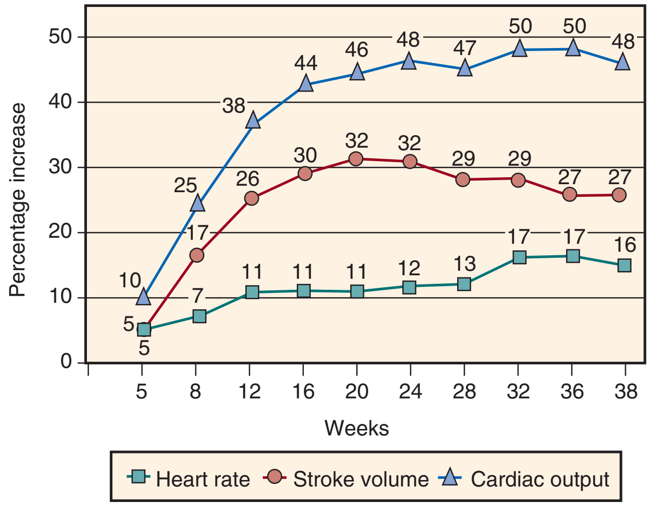

Fig. 48.2 Hemodynamic changes in pregnancy. Shown are the percentage changes from prepregnancy values in heart rate, stroke volume, and cardiac output measured throughout pregnancy. (Modified from Robson SC, Hunter S, Boys RJ, et al. Serial study of factors influencing changes in cardiac output during human pregnancy. *Am J Physiol.* 1989;256:H1060–H1065.)

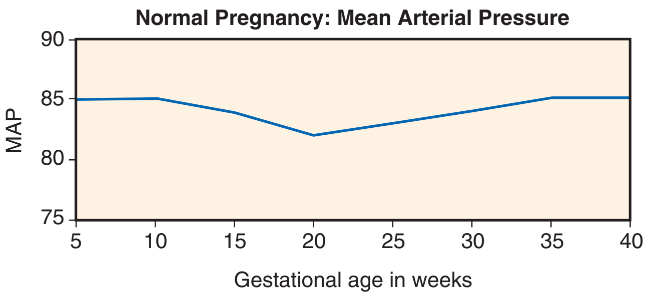

Fig. 48.1 Changes in mean arterial pressure in normal pregnancy. Mean arterial blood pressure (MAP) according to gestational age in weeks in a large representative cohort of pregnant women followed longitudinally. (Adapted from Thadhani R, Ecker JL, Kettyle E, et al. Pulse pressure and risk of preeclampsia: a prospective study. *Obstet Gynecol.* 2001;97:515–520.)

The renin–aldosterone–angiotensin system is activated in pregnancy,[4] leading to renal salt and water retention. Renal interstitial compliance is increased, whereby the renal interstitial pressure remains low despite increased renal interstitial volume. This may contribute to volume retention via an attenuation of the renal pressure natriuretic response.[5] Total body water increases by 6 to 8 liters, leading to both plasma volume and interstitial volume expansion. Thus most women have clinical edema at some point during pregnancy. There is also cumulative retention of about 950 mmol of sodium distributed between the maternal extracellular compartments and the fetus.[6] The plasma volume increases out of proportion to the red blood cell mass, leading to mild physiologic anemia.[7] Plasma volume expansion is followed by increased atrial natriuretic peptide secretion by the late first trimester.[2]

RENAL ADAPTATION TO PREGNANCY

In pregnancy, the kidney length increases by 1–1.5 cm and kidney volume increases by up to 30%.[8] There is physiologic dilatation of the urinary collecting system in approximately 50% of pregnant women, more frequently on the right than the left (Fig. 48.3).[9] These changes may be due to mechanical compression of the ureters between the gravid uterus and the linea terminalis,[10] and the effects of estrogen, progesterone, and prostaglandins on ureteral structure and peristalsis. Hydronephrosis of pregnancy is usually asymptomatic, but abdominal pain, and rarely obstruction, can occur (see the "Obstructive Uropathy and Nephrolithiasis" section).

Glomerular filtration increases by about 40% within weeks of conception, and is maintained at this level until delivery. In early pregnancy, this is mediated primarily by an increase in renal plasma flow (Fig. 48.4). In the second half of pregnancy, renal plasma flow declines toward prepregnancy levels, yet glomerular filtration rate (GFR) remains high, reflecting relatively increased filtration fraction.[11] The maintenance of high GFR despite a fall in renal plasma flow is possible due to decreased capillary oncotic pressure and increased K_f (the product of hydraulic permeability and total surface area available for filtration).[12]

The increase in GFR results in a physiologic decrease in serum creatinine, urea, and uric acid. Normal creatinine clearance in pregnancy rises to 150–200 mL/min, and average

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Fig. 48.3 Physiologic hydronephrosis. Renal ultrasound of the right kidney at 37 weeks' gestation, showing grade 3 (>15 mm) physiologic hydronephrosis of pregnancy. (From Faúndes A, Brícola-Filho M, Pinto e Silva JL. Dilatation of the urinary tract during pregnancy: proposal of a curve of maximal caliceal diameter by gestational age. *Am J Obstet Gynecol,* 1998;178:1082–1086.)

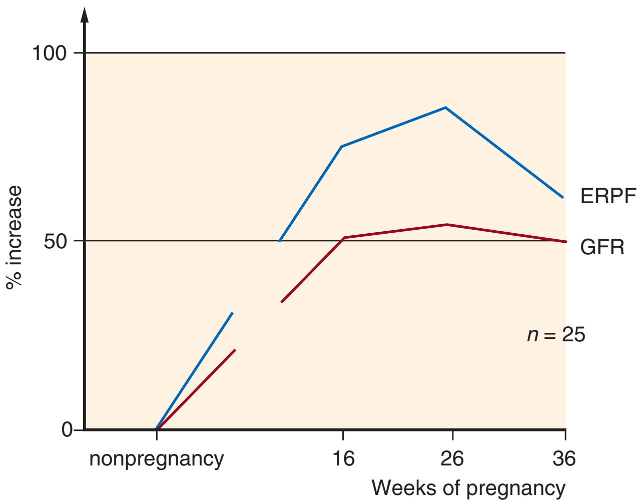

Fig. 48.4 Effect of pregnancy on glomerular filtration rate (GFR) and effective renal plasma flow (ERPF). Renal plasma flow rises out of proportion to the GFR, leading to a decrease in filtration fraction. Both GFR and ERPF peak at midgestation, at approximately 50% and 80% above prepregnancy levels, respectively, and decrease slightly toward term. (From Davison JM. Overview: kidney function in pregnant women. *Am J Kidney Dis.* 1987;9:248.)

serum creatinine falls from 0.8 to 0.5–0.6 mg/dL. Hence a serum creatinine of 1.0 mg/dL, which would be considered normal in a nonpregnant individual, reflects renal impairment in a pregnant woman. Similarly, urea (measured as blood urea nitrogen) falls from an average of 13 mg/dL in the nonpregnant state to approximately 8–10 mg/dL.

Urinary protein excretion rises in normal pregnancy from the nonpregnant level of about 100 to about 180–200 mg/day in the third trimester and occasionally up to 300 mg/day prior to delivery. In women carrying a multiple gestation, physiological proteinuria is even greater, with almost half of women excreting >300 mg/day.[13] This may result in positive urine dipstick testing for protein, particularly on a concentrated urine sample. The increased proteinuria in pregnancy results from a combination of increased GFR, increased permeability of the glomerular basement membrane, and reduced tubular reabsorption of filtered protein.[12] Likewise, women with preexisting proteinuric renal disease appear to have an exacerbation of proteinuria in the second and third trimesters, which is more exaggerated than that which would be expected from the increased GFR alone.[14] (see the "Chronic Kidney Disease in Pregnancy" section).

Serum uric acid declines in early pregnancy because of the rise in GFR, reaching a nadir of 2.0 to 3.0 mg/dL by 22–24 weeks.[15] Thereafter, the uric acid level begins to rise, reaching nonpregnant levels by term. The late gestational rise in uric acid levels is attributed to increased renal tubular absorption of urate.

Pregnancy is characterized by several changes in renal tubular function. Due to the large increase in GFR, glomerular tubular balance requires a concomitant increase in tubular solute reabsorption in order to avoid excessive renal losses. The kidney achieves this balance flawlessly, and sodium balance is maintained normally: pregnant women have normal excretion of an exogenous solute load, and appropriately conserve sodium when intake is restricted.[16]

The ability to excrete a water load is also normally maintained, albeit at a lower osmotic set point. The osmotic threshold for stimulation of both antidiuretic hormone release and thirst is decreased, by a mechanism that appears to be mediated by human chorionic gonadotropin (hCG)[17] and relaxin.[18] This results in mild hyponatremia: The serum sodium typically falls by 4–5 mM/L below nonpregnancy levels. Animal studies suggest that increased aquaporin 2 expression in the collecting tubule may contribute to this effect.[19]

Mild glycosuria and aminoaciduria can occur in normal pregnancy. This occurs in the absence of hyperglycemia or renal disease. These are thought to be due to a combination of the increased filtered load of glucose and amino acids and less efficient tubular reabsorption.

RESPIRATORY ALKALOSIS OF PREGNANCY

Minute ventilation begins to rise by the end of the first trimester, and continues to increase until term. Progesterone mediates this response by direct stimulation of respiratory drive and by increasing sensitivity of the respiratory center to CO_2.[20] This results in a mild respiratory alkalosis—partial pressure of carbon dioxide (pCO_2) falls to approximately 27–32 mm Hg—and a compensatory increase in renal excretion of bicarbonate. This large increase in minute ventilation allows maintenance of high-normal partial pressure of oxygen (pO_2) despite the 20%–33% increase in oxygen consumption in pregnancy.

DIABETES INSIPIDUS OF PREGNANCY

For reasons that are obscure, circulating levels of vasopressinase, an enzyme that hydrolyzes arginine vasopressin, are increased during normal pregnancy. The gene product mediating placental vasopressinase activity was recently characterized as a novel placental leucine aminopeptidase.[21] Occasionally, this increase is so pronounced that circulating antidiuretic hormone (arginine vasopressin) disappears,

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resulting in the polyuria and polydipsia of diabetes insipidus. This syndrome of transient diabetes insipidus presents during the second trimester and disappears after delivery.[22] It is important to recognize this entity because affected women may become dangerously hypernatremic, especially with cesarean section using general anesthesia and/or water restriction in the delivery room. The polyuria can be controlled by the administration of deamino-8-D-arginine vasopressin, which is not destroyed by vasopressinase.[23]

MECHANISM OF VASODILATION IN PREGNANCY

The mechanisms mediating the widespread pregnancy-induced decrease in vascular tone are not fully understood. The fall in SVR is only partially attributable to the presence of the low-resistance circulation in the pregnant uterus, as BP and SVR are noted to fall before this system is well developed. Reduced vascular responsiveness to vasopressors such as angiotensin-2 (AT_2), norepinephrine, and vasopressin in pregnancy is well-documented (Fig. 48.5).[24] Altered vascular receptor expression may contribute to vascular relaxation in pregnancy. For example, animal studies suggest that gestational upregulation of the AT_2 receptor, which produces vasodilation rather than vasoconstriction when stimulated by AT_2, contributes to the fall in BP with pregnancy.[25,26] Multiple hormones and signaling pathways, including estrogen, progesterone, relaxin, and prostaglandins, also contribute to the systemic vasodilatory response.

Pregnancy differs fundamentally from other conditions of peripheral vasodilation, such as sepsis, cirrhosis, and high-output congestive heart failure, all of which are characterized by increased, rather than decreased, renal vascular resistance. This suggests that in pregnancy there is a specific renal vasodilating effect that overrides vasoconstricting factors such as renin–angiotensin–aldosterone activation. Recent advances—mostly from studies of pregnant rats—have suggested that the hormone relaxin is central to this global vasodilatory response, and specifically to the increase in glomerular filtration and renal blood flow.[27] Relaxin is a 6-kDa peptide hormone first isolated from pregnant serum in the 1920s and noted to produce relaxation of the pelvic ligaments.[28] Relaxin is released predominantly from the corpus luteum and rises early in gestation in response to hCG. Gestational renal hyperfiltration and vasodilation were completely abolished in pregnant rats administered relaxin neutralizing antibodies or lacking a functional corpus luteum, suggesting a critical role for relaxin in mediating the renal circulatory changes during pregnancy.[29] Relaxin acts by upregulating endothelin and nitric oxide production in the renal circulation, leading to generalized renal vasodilation, decreased renal afferent and efferent arteriolar resistance, and a subsequent increase in renal blood flow and GFR.[27]

The low-resistance, high-flow circulation of the fetoplacental unit also contributes to the low SVR characteristic of the second and third trimesters of pregnancy. During placental development, the high-resistance uterine arteries are transformed into larger-caliber capacitance vessels (Fig. 48.6). This transformation appears to be driven by invasion of the maternal spiral arteries by fetal-derived cytotrophoblasts, which transform from an epithelial to an endothelial phenotype as they replace the endothelium of the maternal spiral arteries.[30] The mechanisms governing this process, termed "pseudovasculogenesis," are still being elucidated. Angiogenic factors, such as vascular endothelial growth factor (VEGF) and angiopoietins, have a complex spatial and temporal expression in developing placenta, and these factors may be involved in placental vascular development.[31–33]

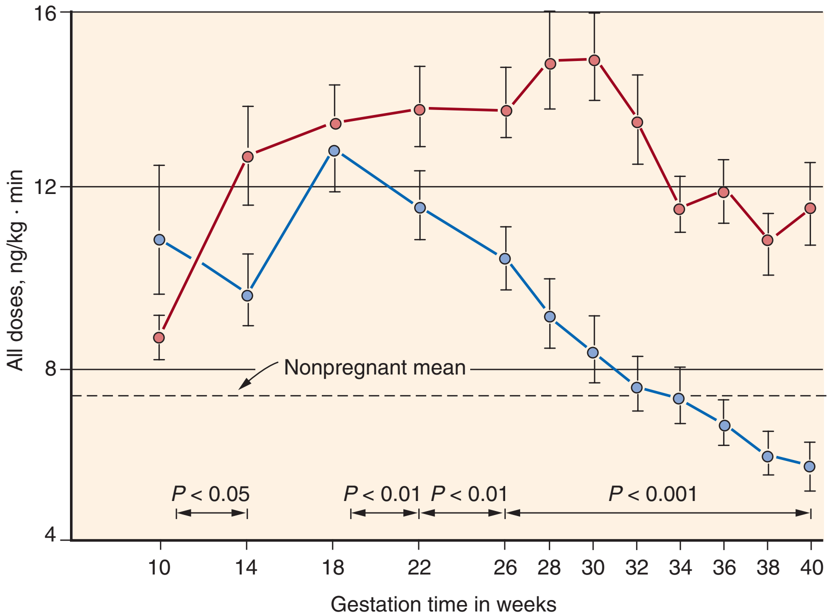

Fig. 48.5 Effect of pregnancy on sensitivity to the pressor effects of angiotensin II. The ordinate displays the dose of angiotensin II needed to raise diastolic blood pressure by 20 mm Hg. In normal pregnancy (*red circles; N* = 120), a higher dose was required than for nonpregnant women *(dashed line).* In women in whom preeclampsia ultimately developed (*blue circles; N* = 72), insensitivity to angiotensin II was lost beginning in mid–second trimester. (From Gant NF, Daley GL, Chand S, et al. A study of angiotensin II pressor response throughout primigravid pregnancy. *J Clin Invest.* 1973;52:2682–2689, by copyright permission of the American Society for Clinical Investigation.)

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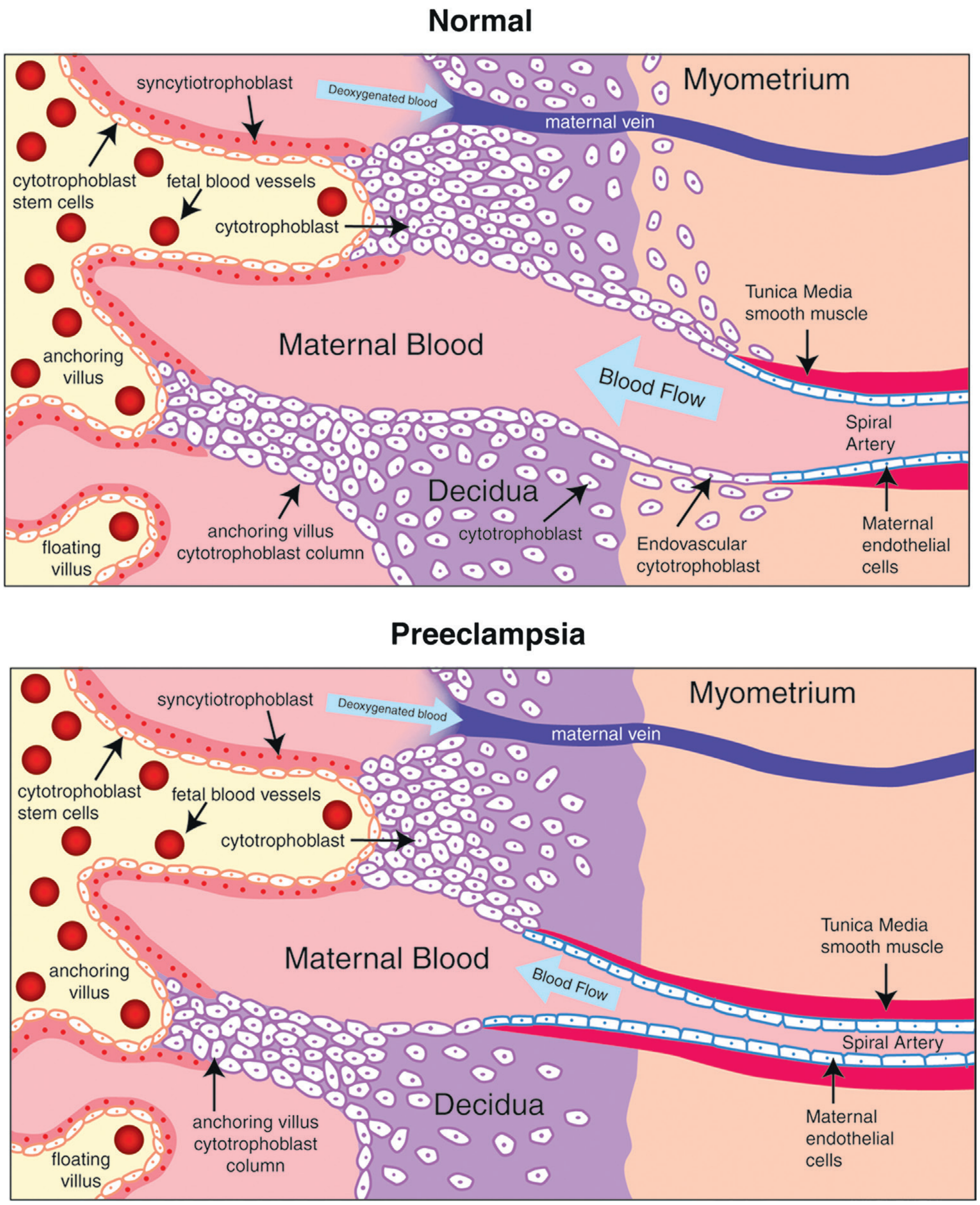

Fig. 48.6 Placentation in normal and preeclamptic pregnancies. In normal placental development *(upper panel)*, invasive cytotrophoblasts of fetal origin invade the maternal spiral arteries, transforming them from small-caliber resistance vessels to high-caliber capacitance vessels capable of providing placental perfusion adequate to sustain the growing fetus. During the process of vascular invasion, the cytotrophoblasts differentiate from an epithelial phenotype to an endothelial phenotype, a process referred to as "pseudovasculogenesis" or "vascular mimicry." In preeclampsia *(lower panel)*, cytotrophoblasts fail to adopt an invasive endothelial phenotype. Instead, invasion of the spiral arteries is shallow, and they remain small-caliber resistance vessels. (From Lam C, Kim KH, Karumanchi SA. Circulating angiogenic factors in the pathogenesis and prediction of preeclampsia. *Hypertension* 2005;46:1077–1085.)

Increased skin capillary density in pregnancy[34] suggests that angiogenic factors may be acting systemically, as well as locally in the placenta. Dysregulation of these angiogenic factors may contribute to disorders of placental vascular development, such as preeclampsia. This is more fully discussed in the next section, "Pathogenesis of Preeclampsia."

PREECLAMPSIA AND HELLP SYNDROME

Preeclampsia is a systemic syndrome that is specific to pregnancy, characterized by the new onset of hypertension and proteinuria after 20 weeks' gestation. Preeclampsia affects approximately 5% of pregnancies worldwide.[35] Despite many advances in our understanding of the pathophysiology of preeclampsia, delivery of the neonate remains the only definitive treatment. Hence preeclampsia is still a leading cause of preterm birth and consequent neonatal morbidity and mortality in the developed world. Maternal mortality from preeclampsia is avoidable with appropriate prenatal and peripartum care, and is now uncommon in developed countries.[36] Globally, where access to safe, emergent delivery is less readily available, preeclampsia continues to claim the lives of over 60,000 mothers every year.[37]

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EPIDEMIOLOGY AND RISK FACTORS

The incidence of preeclampsia varies among populations. Most cases of preeclampsia occur in healthy nulliparous women in whom the incidence of preeclampsia has been reported as high as 7.5%.[38] While classically a disorder of first pregnancies, multiparous women who are pregnant with a new partner appear to have an elevated preeclampsia risk similar to that of nulliparous women.[39] This effect may be due to increased interpregnancy interval rather than the change in partner per se.[40]

Although most cases of preeclampsia occur in the absence of a family history, the presence of preeclampsia in a first-degree relative increases a woman's risk of severe preeclampsia twofold to fourfold,[41] suggesting a genetic contribution to the disease. Several large genome-wide scans seeking a specific linkage to preeclampsia have been fairly discordant and disappointing, with significant logarithm of odds (LOD) scores in isolated Finnish (2p25, 9p13)[42] and Icelandic (2p12)[43] populations. Specific genetic mutations consistent with these loci have remained elusive.

Several medical conditions are associated with increased preeclampsia risk, including chronic hypertension, diabetes mellitus, renal disease, obesity, and antiphospholipid antibody syndrome (Table 48.2).[44] Women with preeclampsia in a prior pregnancy have a high risk of preeclampsia in subsequent pregnancies. Women with early preterm preeclampsia have the highest rate of recurrence.[45]

Adults who were born with low birth weights are at increased risk of pregnancy complications including preeclampsia and gestational hypertension.[46] Conditions associated with increased placental mass, such as multifetal gestations and hydatidiform mole, also are associated with increased preeclampsia risk. Trisomy 13 is associated with a high risk of preeclampsia.[47,48] Use of assisted reproductive technologies has also emerged as an important risk factor for preeclampsia,[49] particularly those methods that involve oocyte donation, such as in vitro fertilization.[50] Although none of these risk factors is fully understood, they have provided insights into pathogenesis.

Several putative risk factors remain controversial. Teen pregnancy has been identified as a risk factor in some studies,[51,52] but this has not been confirmed in a meta-analysis and systematic review.[53] Congenital or acquired thrombophilia is associated with preeclampsia in some[54,55] but not all[56,57] studies. Racial differences in the incidence and severity of preeclampsia have been difficult to assess due to confounding by socioeconomic and cultural factors. Although population-based studies have reported a higher rate of preeclampsia among black women,[58,59] these findings have not been confirmed in studies confined to healthy, nulliparous women.[60,61] This suggests that the increased preeclampsia incidence noted in some studies may be attributable to the higher rate of chronic hypertension in African Americans, as chronic hypertension is itself a strong risk factor for preeclampsia.[62] Black women with preeclampsia also have a higher case-mortality rate,[63] which may be due to more severe disease or to deficiencies in prenatal care. In Hispanics, the incidence of preeclampsia appears to be increased with a concomitant decrease in risk of gestational hypertension (for a discussion on new onset of hypertension without proteinuria, after 20 weeks' gestation, see Chronic Hypertension and Gestational Hypertension).[64]

The possibility that infection may contribute to preeclampsia risk has continued to have sporadic support. A systematic review and meta-analysis of 49 studies reported a small but significant association between preeclampsia and urinary tract infections [odds ratio (OR) 1.57] and periodontal disease (OR 1.76), but no association with other infections, including human immunodeficiency virus (HIV), cytomegalovirus, *Chlamydia*, and malaria.[65] Subsequent studies have supported the association between preeclampsia and urinary tract infection[66] and periodontal disease.[67] It is hypothesized that maternal inflammation predisposes to preeclampsia by causing endothelial stress.

Table 48.2 Risk Factors for Preeclampsia

Risk Factor	Pooled Relative Risk (95% CI)
Antiphospholipid antibody syndrome	2.8 (1.8–4.3)[44]
Chronic kidney disease	1.8 (1.5–2.1)[44]
Prior preeclampsia	8.4 (7.1–9.9)[44]
Nulliparity	2.1 (1.9–2.4)[44]
Chronic hypertension	5.1 (4.0–6.5)[44]
Pregestational diabetes mellitus	3.7 (3.1–4.3)[44]
Multiple gestations	2.9 (2.6–3.1)[44]
Strong family history of cardiovascular disease (heart disease/stroke in two or more first-degree relatives)	3.2 (1.4–7.7)[474]
Systemic lupus erythematosus	2.5 (1.0–6.3)[44]
Obesity (prepregnancy BMI > 30 kg/m^2)	2.8 (2.6–3.1)[44]
Family history of preeclampsia	2.4 (1.8–3.6)[41]
Advanced maternal age (>40 years)	1.5 (1.2–2.0)[44]
Excessive gestational weight gain (>35 lbs)	1.9 (1.7–2.0)[273]
Assisted reproductive technology	1.8 (1.6–2.1)[44]
Previous episode of acute kidney injury	5.9 (3.6–9.7)[475]

BMI, Body mass index; *CI,* confidence interval.

PREECLAMPSIA: DIAGNOSIS AND CLINICAL FEATURES

For many years, the diagnosis of preeclampsia required the new onset of both hypertension and proteinuria after 20 weeks' gestation. In 2013, the Task Force of the American College of Obstetrics and Gynecology published updated criteria for the diagnosis of preeclampsia (Table 48.3) that allow preeclampsia to be diagnosed in the absence of proteinuria if one or more features of severe preeclampsia are present.[68] These guidelines help to distinguish preeclampsia from other hypertensive disorders of pregnancy such as chronic or gestational hypertension. The diagnosis of preeclampsia in women with chronic hypertension and/or underlying proteinuric renal disease on clinical criteria alone remains challenging.

HYPERTENSION

For the diagnosis of preeclampsia, hypertension is defined as a systolic blood pressure (SBP) of ≥140 mm Hg or a diastolic

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Table 48.3 Diagnostic Criteria for Preeclampsia

Diagnostic Criteria for Preeclampsia	
Hypertension	≥140 mm Hg systolic or ≥90 mm Hg diastolic after 20 weeks of gestation on two occasions at least 4 hours apart in a woman with a previously normal blood pressure *OR* With blood pressures ≥160 mm Hg systolic or ≥105 mm Hg diastolic, hypertension can be confirmed within a short interval (minutes) to facilitate timely antihypertensive therapy
AND	
Proteinuria	≥300 mg/24 h (or this amount extrapolated from a timed collection) *OR* Protein-to-creatinine ratio ≥ 0.3 mg protein/mg creatinine *OR* Dipstick 2+ (used only if other quantitative methods not available)
OR in the absence of proteinuria, new-onset hypertension with the new onset of any of the following:	
Thrombocytopenia	≤100,000 platelets/mL
Renal insufficiency	Serum creatinine concentrations >1.1 mg/dL or a doubling of the serum creatinine concentration in the absence of other renal disease
Impaired liver function	Elevated blood concentrations of liver transaminases to twice normal
Pulmonary edema	
Cerebral or visual symptoms	
Diagnostic Criteria for Superimposed Preeclampsia	
Hypertension	A sudden increase in blood pressure in a woman with chronic hypertension that was previously well controlled or escalation of antihypertensive medications to control blood pressure
OR	
Proteinuria	New onset of proteinuria in a woman with chronic hypertension or a sudden increase in proteinuria in a woman with known proteinuria before or in early pregnancy

Adapted from American College of Obstetricians and Gynecologists, Task Force on Hypertension in Pregnancy. Hypertension in pregnancy. Report of the American College of Obstetricians and Gynecologists' Task Force on Hypertension in Pregnancy. *Obstet Gynecol.* 2013;122:1122–1131 and ACOG Practice Bulletin No 202: Gestational Hypertension and Preeclampsia. *Obstet Gynecol.* 2019;133(1):e1–e25.

blood pressure (DBP) of ≥90 mm Hg after 20 weeks' gestation in a woman with previously normal BP.[68] Hypertension should be confirmed by two separate measurements at least 4 hours apart. The severity of hypertension in preeclampsia can vary widely, from mild BP elevations easily managed with antihypertensive medication to severe hypertension associated with headache and visual changes resistant to multiple medications. The latter situation can often herald seizures (eclampsia) and is an indication for urgent delivery. Medical management of hypertension in preeclampsia is discussed in the next section.

PROTEINURIA

Proteinuria is a hallmark of preeclampsia. For the diagnosis of preeclampsia, proteinuria is defined as >300 mg protein in a 24-hour urine collection or a urine protein-to-creatinine (P:C) ratio >0.3 mg/mg, or urine dipstick of ≥2+ (if quantitative methods are unavailable). However, preeclampsia can be diagnosed even in the absence of proteinuria, if other severe features are present (Table 48.3).

Routine obstetric care includes dipstick protein testing of a random voided urine sample at each prenatal visit – a screening method that has been shown to have a high rate of false-positive and false-negative results when compared with 24-hour urine protein measurement.[69] However, the 24-hour urine collection for proteinuria is cumbersome for the patient, often inaccurate due to undercollection,[70] and the availability of results is delayed for at least 24 hours, while the collection is being completed. The urinary P:C ratio has become the preferred method for quantification of proteinuria in the nonpregnant population. A meta-analysis showed a pooled sensitivity of 84% and specificity of 76% using P:C ratio cutoff of >0.3, as compared with the gold standard of 24-hour urine protein excretion >300 mg/day.[71] Hence it is reasonable to use the urine P:C ratio for the diagnosis of preeclampsia, with 24-hour collection undertaken when the result is equivocal. A 12-hour urine collection showing >150 mg protein may also be used as a surrogate for 24-hour urine collection, according to a meta-analysis of several studies.[72]

The degree of proteinuria in preeclampsia can range widely, from minimal to nephrotic range. However, the degree of proteinuria is a poor predictor of adverse maternal and fetal outcomes,[73] thus heavy proteinuria alone is not an indication for urgent delivery. New-onset proteinuria is particularly useful among women with chronic hypertension to diagnose superimposed preeclampsia (Table 48.3). However, among women with underlying proteinuria, other signs of preeclampsia such as elevated transaminases, thrombocytopenia, or cerebral signs/symptoms are more useful to diagnose superimposed preeclampsia.[68]

EDEMA

Although edema was historically part of the diagnostic triad for preeclampsia, it is also recognized to be a feature of normal pregnancy, diminishing its usefulness as a specific pathologic sign. Still, the sudden onset of severe edema—especially edema of the hands and face—can be an important presenting symptom in this otherwise insidious disease and should prompt evaluation.

URIC ACID

Serum uric acid is elevated in most women with preeclampsia primarily as a result of enhanced tubular urate reabsorption. It has been suggested that hyperuricemia may contribute to the pathogenesis of preeclampsia by inducing endothelial dysfunction[74] or by impairing trophoblast invasiveness, a key

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element in placental vascular remodeling.[75] Serum uric acid levels are correlated with the presence and severity of preeclampsia and with adverse pregnancy outcomes,[76] even in gestational hypertension without proteinuria.[77] Unfortunately, uric acid is of limited clinical utility in either distinguishing preeclampsia from other hypertensive disorders of pregnancy or as a clinical predictor of adverse outcomes.[78–80] One clinical scenario where serum uric acid may be useful is in the diagnosis of preeclampsia in women with chronic kidney disease (CKD), in whom the usual diagnostic criteria of new-onset hypertension and proteinuria are often impossible to apply. In such patients, a serum uric acid level >5.5 mg/dL in the presence of stable renal function might suggest superimposed preeclampsia.

CLINICAL FEATURES OF SEVERE PREECLAMPSIA

Several clinical and laboratory findings suggest severe or progressive disease, and should prompt consideration of immediate delivery.[68] Oliguria (<500 mL urine in 24 hours) is usually transient; acute kidney injury (AKI), though uncommon, can occur. Persistent headache or visual disturbances can be a prodrome to seizures. Pulmonary edema complicates 2%–3% of severe preeclampsia,[39] and can lead to respiratory failure. Epigastric or right upper quadrant pain may be associated with liver injury. Elevated liver enzymes and thrombocytopenia can occur alone or as part of the HELLP (*h*emolytic anemia, *e*levated *l*iver enzymes, and *l*ow *p*latelets) syndrome (see later).

ECLAMPSIA

Seizures complicate approximately 2% of preeclampsia cases in the United States.[81] Although eclampsia most often occurs in the setting of hypertension and proteinuria, it can occur without these warning signs. Up to one-third of eclampsia occurs postpartum, sometimes days to weeks after delivery.[82] Late postpartum preeclampsia in particular is often a difficult and potentially missed diagnosis, often seen by nonobstetricians in the emergency room. Radiologic imaging by head computed tomography (CT) or magnetic resonance imaging (MRI) is usually not indicated when the diagnosis is apparent, but typically shows vasogenic edema, predominantly in the subcortical white matter of the parietooccipital lobes (see "Etiology of Pathogenesis of Preeclampsia: Cerebral Changes"). Women who have had eclampsia may have subtle long-term impairment in cognitive function.[83]

HELLP SYNDROME

"HELLP" is an acronym for the syndrome of *h*emolytic anemia, *e*levated *l*iver enzymes, and *l*ow *p*latelets. There remains considerable variability regarding diagnostic criteria for the HELLP syndrome in the medical literature. The HELLP syndrome is a severe form of preeclampsia, which can occur in the absence of proteinuria. As its name suggests, a diagnosis of HELLP syndrome is suggested by evidence of hemolytic anemia, thrombocytopenia (platelet count < 100,000/μL), and impaired liver function (liver enzyme levels more than twice the upper limit of normal). This syndrome is sometimes challenging to distinguish from thrombotic thrombocytopenic purpura (TTP), hemolytic uremic syndrome (HUS), and acute fatty liver of pregnancy (AFLP), which can present similarly (Table 48.4). The HELLP syndrome is associated with particularly high rates of adverse maternal and neonatal adverse outcomes, including eclampsia (affecting 6% of cases), placental abruption (10%), acute renal failure (5%), disseminated intravascular coagulation (8%), pulmonary edema (10%),[84] and (rarely) hepatic hemorrhage and rupture.[85]

Table 48.4 Comparison of Clinical and Laboratory Characteristics, Effect on Delivery, and Management of HELLP, HUS/TTP, and AFLP

	HUS/TTP	HELLP	AFLP
Clinical Characteristic:			
Hemolytic anemia	+++	++	±
Thrombocytopenia	+++	++	±
Coagulopathy	–	±	+
CNS symptoms	++	±	±
Renal failure	+++	+	++
Hypertension	±	+++	±
Proteinuria	±	++	±
Elevated AST	±	++	+++
Elevated bilirubin	++	+	+++
Anemia	++	+	±
Ammonia	Normal	Normal	High
Effect of delivery on disease	None	Recovery	Recovery
Management	Plasma exchange	Supportive care, delivery	Supportive care, delivery

AFLP, Acute fatty liver of pregnancy; *AST,* aspartate aminotransferase; *CNS,* central nervous system; *HELLP, h*emolytic anemia, elevated *l*iver enzymes, and *l*ow *p*latelets; *HUS/TTP,* hemolytic uremic syndrome/thrombotic thrombocytopenic purpura; –, absent; ±, present or absent; +, mild; ++, moderate; +++, severe.

Data derived from Allford SL, Hunt BJ, Rose P, et al. Guidelines on the diagnosis and management of the thrombotic microangiopathic haemolytic anaemias. *Br J Haematol*. 2003;120:556–573; Egerman RS, Sibai BM. Imitators of preeclampsia and eclampsia. *Clin Obstet Gynecol*. 1999; 42:551–562; Stella CL, Dacus J, Guzman E, et al. The diagnostic dilemma of thrombotic thrombocytopenic purpura/hemolytic uremic syndrome in the obstetric triage and emergency department: lessons from 4 tertiary hospitals. *Am J Obstet Gynecol.* 2009;200:381–386.

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MATERNAL AND NEONATAL MORTALITY

Approximately 300,000 women die in childbirth each year worldwide, and hypertensive disorders of pregnancy are estimated to account for 10% to 20% of these deaths.[37,86] In the United States, the rate of severe preeclampsia has been increasing since the 1980s,[51] and preeclampsia and eclampsia account for 16%–20% of all pregnancy-related maternal mortality.[63,87] Maternal death is most often due to eclampsia, cerebral hemorrhage, renal failure, hepatic failure, pulmonary edema, and the HELLP syndrome. Most preventable errors in preeclampsia management leading to maternal death involve inattention to BP control and signs of pulmonary edema.[87] Risk of death in preeclampsia is increased for women with little or no prenatal care, women of black race, those over 35 years of age, and those with early-onset preeclampsia.[63] Adverse maternal outcomes can often be avoided with timely delivery; hence in the developed world the burden of morbidity and mortality falls on the neonate.

Worldwide, preeclampsia is associated with a perinatal and neonatal mortality rate of 10%[88]; as with maternal mortality, the risk of neonatal mortality increases substantially for preeclampsia presenting earlier in gestation. Neonatal death is most commonly due to iatrogenic prematurity undertaken to preserve the health of the mother. In addition, fetal growth restriction can occur, likely as a result of impaired uteroplacental blood flow or placental infarction. Oligohydramnios and placental abruption are less common complications.

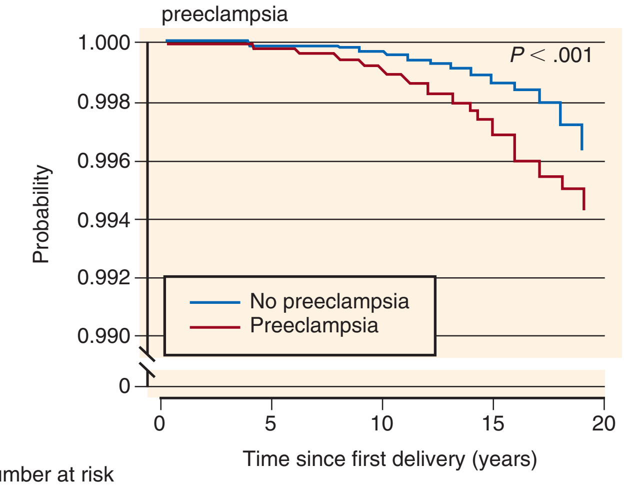
POSTPARTUM RECOVERY

Generally, preeclampsia begins to remit soon after delivery of the fetus and placenta and complete recovery is the rule. However, normalization of BP and proteinuria often takes days to weeks.[89] Postpartum monitoring is important because, as discussed, eclampsia can occur after delivery.

LONG-TERM CARDIOVASCULAR AND RENAL OUTCOMES

Previously, women with preeclampsia were reassured that the syndrome remits completely after delivery, with no long-term consequences aside from increased preeclampsia risk in future pregnancies. Epidemiologic studies have refuted this claim.[90,91] Fifty percent of women with preeclampsia develop hypertension later in life. As early as 1 year after the affected pregnancy, women with preeclampsia have an increased risk of hypertension, obesity, hypercholesterolemia, microalbuminuria, and new-onset diabetes mellitus, when compared with control women who have not had preeclampsia.[92–94] Relative risk (RR) of ischemic heart disease, stroke, cardiomyopathy, and cardiovascular mortality are more than doubled in women who have had preeclampsia.[90,91,95] (Fig. 48.7) Severe preeclampsia, recurrent preeclampsia, preeclampsia with preterm birth, and preeclampsia with intrauterine growth restriction (IUGR) are most strongly associated with adverse cardiovascular outcomes. The American Heart Association Guidelines include a history of preeclampsia as a risk factor for cardiovascular disease in women.[96]

Preeclampsia, especially in association with low neonatal birth weight, also carries an increased risk of later maternal kidney disease requiring a kidney biopsy.[97] A large Norwegian study by the same authors, using birth and renal registry data on >570,000 women, showed that preeclampsia increases the risk of subsequent end-stage renal disease (ESRD) by almost fivefold.[98] This finding has subsequently been confirmed in other populations.[99] Familial aggregation of risk factors does not seem to explain increased ESRD risk after preeclampsia.[100] Although it appears that preeclampsia is associated with increased risk of subsequent ESRD, the absolute risk is low (e.g., 4.72 per 10,000 person-years in the Taiwanese study).[99]



Number at risk

No preeclampsia	106,509	106,364	106,111	84,487
Preeclampsia	22,781	22,744	22,669	18,409

Fig. 48.7 Preeclampsia increases the risk for cardiovascular disease later in life. Kaplan–Meier plot of the cumulative probability of survival without admission to the hospital for ischemic heart disease or death from ischemic heart disease in women with and without a history of preeclampsia. (From Smith GC, Pell JP, Walsh D. Pregnancy complications and maternal risk of ischaemic heart disease: a retrospective cohort study of 129,290 births. *Lancet* 2001;357:2002–2006.)

Preeclampsia and cardiovascular disease share many common risk factors, such as chronic hypertension, diabetes, obesity, renal disease, and the metabolic syndrome. Still, the increase in long-term cardiovascular mortality holds even for women who develop preeclampsia in the absence of any overt vascular risk factors. Whether these observations result from vascular damage or persistent endothelial dysfunction caused by preeclampsia, or simply reflect the common risk factors shared by preeclampsia and cardiovascular disease, remains speculative. Regardless of etiology, it is recommended that women who experience preeclampsia, especially with preterm birth or IUGR, receive screening for potentially modifiable cardiovascular and renal disease risk factors (hypertension, diabetes mellitus, hyperlipidemia, obesity) at their postpartum obstetrician visit and yearly thereafter.[68,101,102]

Epidemiologic evidence suggests that low birth weight (with or without preeclampsia) is associated with the development of hypertension, diabetes, cardiovascular disease, and CKD in the offspring of affected pregnancies.[103–106] It has been hypothesized that this may be in part due to low nephron number as is further discussed in Chapter 21.

PATHOGENESIS OF PREECLAMPSIA

THE ROLE OF THE PLACENTA

Observational evidence suggests that the placenta has a central role in preeclampsia. Preeclampsia only occurs in the presence

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of a placenta—though not necessarily a fetus, as in the case of hydatidiform mole—and almost always remits after its delivery. In a case of preeclampsia with extrauterine pregnancy, removal of the fetus alone was not sufficient; symptoms persisted until the placenta was delivered.[107] Severe preeclampsia is associated with pathologic evidence of placental hypoperfusion and ischemia. Findings include acute atherosis, a lesion of diffuse vascular obstruction that includes fibrin deposition, intimal thickening, necrosis, atherosclerosis, and endothelial damage.[108] Infarcts, likely due to occlusion of maternal spiral arteries, are also common. Although these findings are not universal, they appear to be correlated with severity of clinical disease.[109]

Several clinical and laboratory observations strongly support the role of placental ischemia in the pathophysiology of preeclampsia. Abnormal uterine artery Doppler ultrasound, consistent with decreased uteroplacental perfusion, is observed before the clinical onset of preeclampsia.[110] The incidence of preeclampsia is increased twofold to fourfold in women residing at high altitude, implying hypoxia may be a contributing factor.[111] Indeed, global gene expression profiles are similar when comparing hypoxia-treated placental explants, high-altitude placentae, or placentae from preeclamptic pregnancies.[112] Pregnant subjects with sickle cell disease, who often have pathologic evidence of placental ischemia and infarction, have an increased risk for preeclampsia.[113–115] Hypertension and proteinuria can be induced by constriction of uterine blood flow in pregnant primates and other mammals.[116] These observations suggest that placental ischemia may be an important trigger for the maternal syndrome.

However, evidence for a causative role for placental ischemia alone remains circumstantial, and several observations call the hypothesis into question. For example, the animal models based on uterine hypoperfusion fail to induce several of the multiorgan features of preeclampsia, including seizures, elevated liver enzymes, and thrombocytopenia. In most cases of preeclampsia, there is no evidence of growth restriction or fetal intolerance of labor, which are expected consequences of placental ischemia. It may be that the placental ischemic damage that accompanies late-stage preeclampsia may be a secondary event.

PLACENTAL VASCULAR REMODELING

Early in normal placental development, extravillous cytotrophoblasts invade the uterine spiral arteries of the decidua and myometrium (see Fig. 48.6). These invasive fetal cells replace the endothelial layer of the uterine vessels, transforming them from small resistance vessels to flaccid, high-caliber capacitance vessels.[117] This vascular transformation is most dramatic in the myometrial vessels and allows the increase in uterine blood flow needed to sustain the fetus through the pregnancy.[118] In preeclampsia, this transformation is incomplete.[119] Cytotrophoblast invasion of the arteries is limited to the superficial decidua, and the myometrial segments remain narrow and undilated.[120] Fisher et al[121] have shown that in normal placental development, invasive cytotrophoblasts downregulate the expression of adhesion molecules characteristic of their epithelial cell origin and adopt an endothelial cell-surface adhesion phenotype, a process dubbed pseudovasculogenesis. In preeclampsia, cytotrophoblasts do not undergo this switching of cell-surface integrins and adhesion molecules, and fail to adequately invade the myometrial spiral arteries.

The factors that regulate this process are just beginning to be elucidated. Hypoxia-inducible factor-1 (HIF-1) activity is increased in preeclampsia and HIF-1 target genes such as encoding transforming growth factor β-3 *TGFB3,* may block cytotrophoblast invasion.[122] Invasive cytotrophoblasts express several angiogenic factors and receptors, also regulated by HIF, including VEGF, placental growth factor (PlGF), and VEGF receptor 1 (VEGFR-1 orFlt1); expression of these proteins by immunolocalization is altered in preeclampsia.[123] A genetic study identified polymorphisms in *STOX1*, a paternally imprinted gene and member of the winged helix gene family, in a Dutch preeclampsia cohort.[124] The authors hypothesized that loss-of-function mutations in this gene could result in defective polyploidization of extravillous trophoblast, leading to loss of cytotrophoblast invasion. However, a subsequent cohort failed to confirm an association between preeclampsia and STOX polymorphisms.[125] More work is needed to uncover the molecular signals governing cytotrophoblast invasion early in placentation, defects in which may underlie the early stages of preeclampsia.

> ***Clinical Relevance***
> Inadequate early placental vascular development leads to development of placental ischemia later in pregnancy, culminating in preeclampsia. Dysregulation of angiogenic factors, such as vascular endothelial growth factor (VEGF), placental growth factor, and soluble fms-like tyrosine kinase-1 (sFlt1), may play a role in both early placental development and later stages of clinical preeclampsia.

MATERNAL ENDOTHELIAL DYSFUNCTION

Although the origins of the preeclampsia syndrome appear to be placental, the target organ is the maternal endothelium. The clinical manifestations of preeclampsia reflect widespread endothelial dysfunction, resulting in vasoconstriction and end-organ ischemia.[126,127] Incubation of endothelial cells with serum from women with preeclampsia results in endothelial dysfunction; hence it has been hypothesized that factors present in maternal serum, likely originating in the placenta, are responsible for the manifestations of the disease (Fig. 48.8).

Dozens of serum markers of endothelial activation are deranged in women with preeclampsia, including von Willebrand antigen, cellular fibronectin, soluble tissue factor, soluble E-selectin, platelet-derived growth factor, and endothelin.[127] C-reactive protein[128] and leptin[129] are increased early in gestation. There is evidence for oxidative stress and platelet activation.[130] Decreased production of prostaglandin I_2, an endothelial-derived prostaglandin, occurs well before the onset of clinical symptoms.[131] Inflammation is often present; for example, there is neutrophil infiltration in the vascular smooth muscle of subcutaneous fat, with increased vascular smooth muscle expression of interleukin-8 (IL-8) and intercellular adhesion molecule 1 (ICAM-1).[132] Several of these aberrations occur well before the onset of symptoms, sup-

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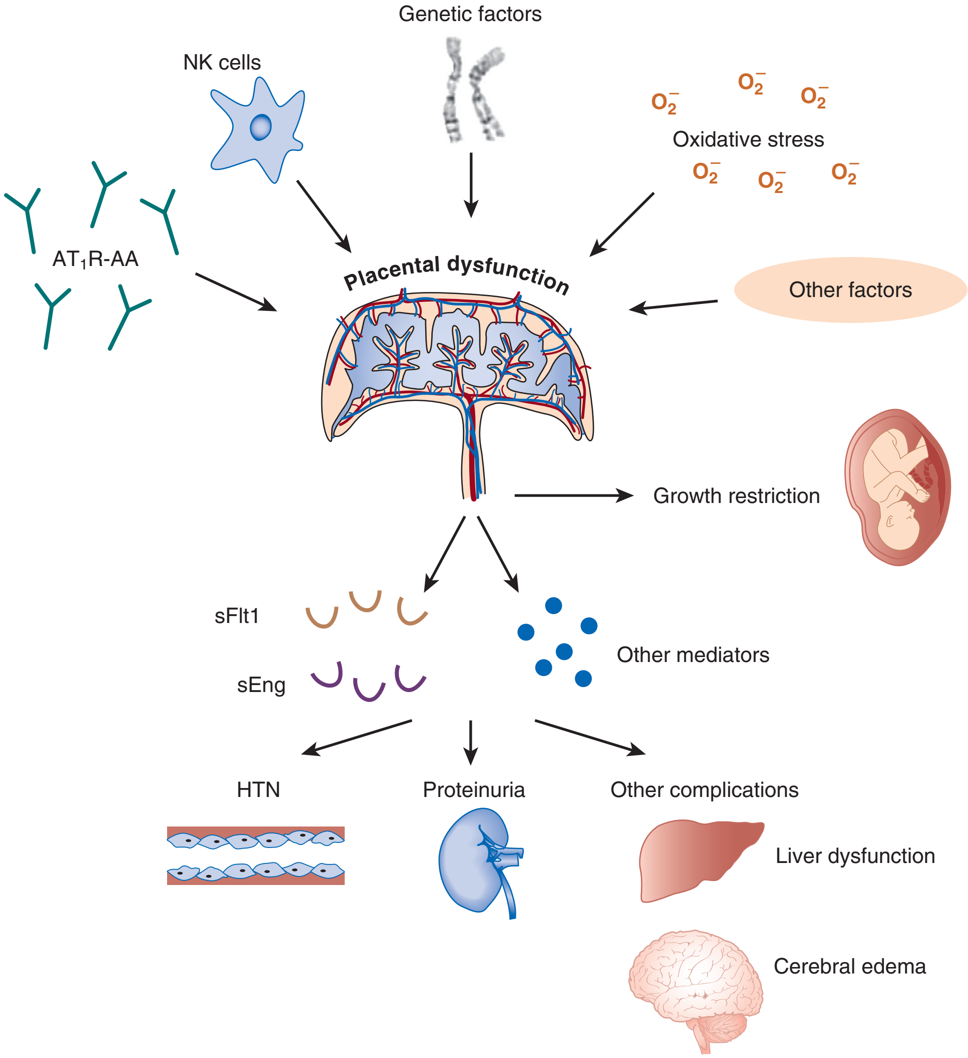

Fig. 48.8 Placental dysfunction and endothelial dysfunction in the pathogenesis of preeclampsia. Placental dysfunction triggered by genetic, immunologic [natural killer (NK) cells and angiotensin receptor autoantibodies (AT_1R-AAs)], and other factors (altered heme oxygenase expression, oxidative stress) plays an early and primary role in the pathogenesis of preeclampsia. The diseased placenta in turn secretes antiangiogenic factors [soluble fms-like tyrosine kinase-1 (sFlt1), soluble endoglin (sEng)] and other toxic mediators into the systemic circulation, causing maternal endothelial dysfunction. Nearly all the manifestations of preeclampsia, including hypertension, proteinuria (glomerular endotheliosis), seizures (cerebral edema), and HELLP (*h*emolytic anemia, *e*levated *l*iver enzymes, and *l*ow *p*latelets) syndrome, can be attributable to vascular endothelial damage secondary to excess circulating antiangiogenic factors. (Adapted from Powe CE, Levine RJ, Karumanchi SA. Preeclampsia, a disease of the maternal endothelium: the role of antiangiogenic factors and implications for later cardiovascular disease. *Circulation* 2011;123: 2856–2869.)

porting the central role of endothelial dysfunction in the pathogenesis of preeclampsia.

HEMODYNAMIC CHANGES

The decreases in peripheral vascular resistance and arterial BP that occur during normal pregnancy are absent or reversed in preeclampsia. SVR is high and cardiac output is low when compared with normal pregnancies.[133] These changes are due to widespread vasoconstriction resulting from endothelial dysfunction. This hypothesis is supported by both in vivo and in vitro evidence. Women with preeclampsia have impaired endothelium-dependent vasorelaxation, which has been noted prospectively prior to onset of hypertension and proteinuria[134] and persists for years after the preeclampsia episode.[135] There is exaggerated sensitivity to vasopressors such as angiotensin II and norepinephrine (see Fig. 48.5).[127] Subtle increases in BP and in pulse pressure are present prior to the onset of overt hypertension and proteinuria, suggesting that arterial compliance is decreased early in the course of the disease.[53,136] Mechanisms underlying endothelial dysfunction are discussed in the next section.

RENAL CHANGES

The pathologic swelling of glomerular endothelial cells in preeclampsia was first described in 1924.[137] Thirty years later, Spargo et al[138] coined the term "glomerular endotheliosis" and characterized ultrastructural changes, including generalized swelling and vacuolization of the endothelial cells and loss of the capillary space (Fig. 48.9). There are deposits of fibrinogen and fibrin within and under the endothelial cells,

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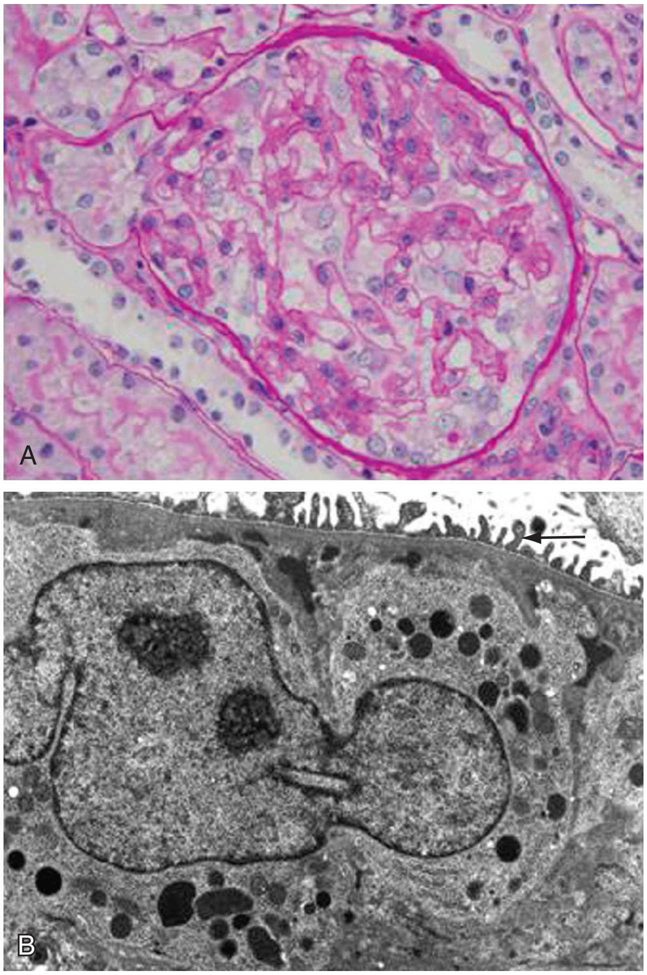

Fig. 48.9 Glomerular endotheliosis. (A) Human preeclamptic glomerulus on light microscopy (periodic acid–Schiff stain). Renal biopsy findings from a 29-year-old woman with twin gestation and severe preeclampsia are shown. Patient's blood pressure was 170/112 mm Hg, and random urine protein-to-creatinine ratio was 9.8. Note the "bloodless" appearance of the glomeruli and absence of the capillary lumen. (Original magnification, ×40.) (B) Electron microscopy of biopsy specimen of the glomerulus from the same patient. Note occlusion of capillary lumen cytoplasm and expansion of the subendothelial space with some electron-dense material. Podocyte cytoplasm shows protein resorption droplets and relatively intact foot processes. (Original magnification, ×1500.) (Courtesy IE Stillman.)

and electron microscopy shows loss of glomerular endothelial fenestrae.[139] The primary injury is specific to endothelial cells: the podocyte foot processes are intact early in disease, a finding atypical of other nephrotic diseases. However, podocyte injury as evidenced by podocyturia has been observed in preeclampsia.[140] Changes in the afferent arteriole, including atrophy of the macula densa and hyperplasia of the juxtaglomerular apparatus, have also been described.[141] Although once considered pathognomonic for preeclampsia, recent studies have shown that mild glomerular endotheliosis also occurs in pregnancy without preeclampsia, especially in gestational hypertension.[142] This suggests that the endothelial dysfunction of preeclampsia may in fact be an exaggeration of a process present toward term in all pregnancies.

Both renal blood flow and GFR are low in preeclampsia as compared with normal pregnancy. Renal blood flow falls as a result of high renal vascular resistance, primarily due to increased afferent arteriolar resistance. GFR falls as a result of both the fall in renal blood flow and a decrease in the ultrafiltration coefficient (K_f), attributed to endotheliosis in the glomerular capillary.[143] Although AKI can occur in preeclampsia, typically proteinuria (with a bland urinary sediment) and renal sodium and water retention are the only renal manifestations of disease.

CEREBRAL CHANGES

Cerebral edema and intracerebral parenchymal hemorrhage are common autopsy findings in women who died from eclampsia. The presence of cerebral edema in eclampsia correlates with markers of endothelial damage but not the severity of hypertension,[144] suggesting that the edema is secondary to endothelial dysfunction rather than a direct result of BP elevation. Findings on head CT and MRI are similar to those seen in hypertensive encephalopathy, with vasogenic cerebral edema and infarctions in the subcortical white matter and adjacent gray matter, predominantly in the parietooccipital lobes.[82] A syndrome that includes these characteristic MRI changes, together with headache, seizures, altered mental status, and hypertension, has been described in patients with acute hypertensive encephalopathy in the setting of renal disease, eclampsia, or immunosuppression.[145] This syndrome, termed reversible posterior leukoencephalopathy, has subsequently been associated with the use of calcineurin inhibitors (CNIs) or antiangiogenic agents for cancer therapy.[146] This latter observation supports the role of innate antiangiogenic factors in the pathophysiology of preeclampsia/eclampsia, as detailed later in this section.

OXIDATIVE STRESS AND INFLAMMATION

Oxidative stress, the presence of reactive oxygen species in excess of antioxidant buffering capacity, is a prominent feature of preeclampsia. Oxidative stress is known to damage proteins, cell membranes, and DNA and is a potential mediator of endothelial dysfunction. It has been hypothesized that in preeclampsia, placental oxidative stress is transferred to the systemic circulation, resulting in oxidative damage to the maternal vascular endothelium.[130] However, the absence of any clinical benefit of antioxidant supplementation in the prevention of preeclampsia suggests that oxidative stress is likely to be a secondary phenomenon in preeclampsia, and not a promising therapeutic target.[147] Circulating placental cytotrophoblast debris and the accompanying inflammation have also been proposed as pathogenic mechanisms to explain the maternal endothelial dysfunction; however, causal evidence for this hypothesis is still lacking.[148]

IMMUNOLOGIC INTOLERANCE

The possibility of immune maladaptation remains an intriguing but unproven theory of the pathogenesis of preeclampsia. Normal placentation requires the development of immune tolerance between the fetus and the mother. The fact that preeclampsia occurs more often in first pregnancies or after a change in partners suggests an etiologic role for abnormal maternal immune response to paternally derived fetal antigens. This could result in failure of fetal cells to successfully invade the maternal vessels during placental vascular development.

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Observational studies suggest that preeclampsia risk increases in cases of exposure to novel paternal antigens – not only in first pregnancies, but also in pregnancies with a new partner[149] and with long interpregnancy interval.[40] Women using contraceptive methods that reduce exposure to sperm have increased preeclampsia incidence.[150] Women impregnated by intracytoplasmic sperm injection in which sperm were surgically obtained (i.e., the woman was never exposed to partner's sperm in intercourse) had a threefold increased risk of preeclampsia compared with intracytoplasmic sperm injection cases where sperm were obtained by ejaculation.[151] Conversely, prior exposure to paternal antigens appears to be protective. The risk of preeclampsia is inversely proportional to the length of cohabitation,[152] and oral tolerance to paternal antigens by oral sex and swallowing is associated with decreased risk.[153] None of these clinical observations have yet provided insights into immunologic triggers or pathogenic links to the paternal syndrome.

On a molecular level, HLA-G expression appears to be abnormal in preeclampsia. HLA-G is normally expressed by invasive extravillous cytotrophoblasts and may play a role in inducing immune tolerance at the maternal–fetal interface. In preeclampsia, HLA-G expression by cytotrophoblasts is reduced or absent,[154] and HLA-G protein concentrations are reduced in maternal serum[155] and in placental tissue.[156] These alterations in HLA-G expression could contribute to the ineffective trophoblast invasion seen in preeclampsia. Decidual natural killer (NK) cells, which promote angiogenesis and are involved in trophoblast invasion, have also been hypothesized to contribute to abnormal placental development seen in the disease.[157,158] Genetic studies have noted that the susceptibility to preeclampsia may be influenced by polymorphisms in killer immunoglobulin receptors (KIRs, present on NK cells) and HLA-C (KIR ligands present on trophoblasts).[159]

ANGIOGENIC IMBALANCE

Overwhelming evidence from epidemiological studies and experimental studies in animals suggests that excess placental production of soluble VEGFR-1, referred to as soluble fms-like tyrosine kinase-1 (sFlt1, or sVEGFR-1), plays a causal role in mediating the signs and symptoms of preeclampsia.[160] sFlt1, a truncated splice variant of the VEGFR Flt1, antagonizes VEGF and PlGF by binding them in the circulation and preventing interaction with their endogenous receptors in the vasculature (Fig. 48.10). sFlt1 inhibits VEGF- and PlGF-mediated angiogenesis and is upregulated in the placenta of women with preeclampsia, resulting in elevated circulating levels.[161] The increase in maternal circulating sFlt1 precedes the onset of clinical disease (Fig. 48.11)[162–164] and is correlated with disease severity.[164,165] Increased circulating sFlt1 is accompanied by decreased circulating free PlGF in serum (Fig. 48.12). In vitro effects of sFlt1 include vasoconstriction and endothelial dysfunction. Exogenous sFlt1 administered to pregnant rats produces a syndrome resembling preeclampsia, including hypertension, proteinuria, and glomerular endotheliosis.[161] The preeclampsia-like syndrome induced by sFlt1 in animals can be rescued by exogenous VEGF or PlGF administration.[166–168] In summary, this work has suggested that sFlt1 is a key pathogenic circulating toxin that mediates the signs and symptoms of preeclampsia (see Fig. 48.8). Recently, several novel isoforms of sFlt1 have been identified in the human placenta; however, the exact role of the various forms in human disease is still being investigated.[169,170]

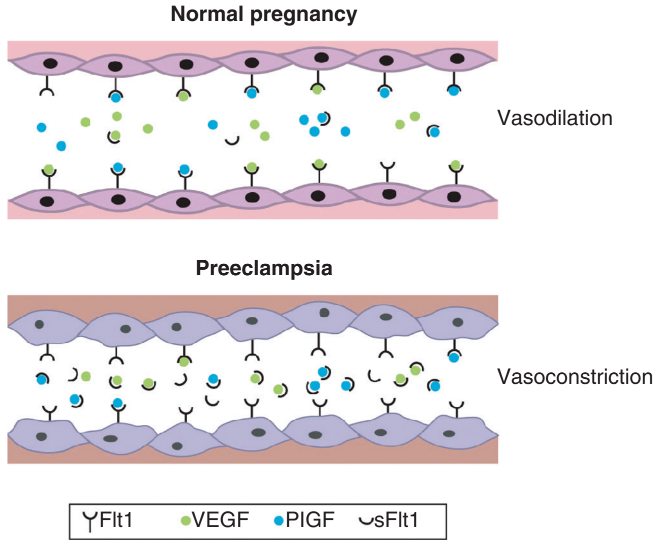

Fig. 48.10 Proposed mechanism of soluble fms-like tyrosine kinase-1 (sFlt1)–induced endothelial dysfunction. sFlt1 protein, derived from alternative splicing of Flt1, lacks the transmembrane and cytoplasmic domains but still has the intact vascular endothelial growth factor (VEGF) and placental growth factor (PlGF) binding extracellular domain. During normal pregnancy, VEGF and PlGF signal through the VEGF receptors (Flt1) and maintain endothelial health. In preeclampsia, excess sFlt1 binds to circulating VEGF and PlGF, thus impairing normal signaling of both VEGF and PlGF through their cell surface receptors. Thus excess sFlt1 leads to maternal endothelial dysfunction. (From Bdolah Y, Sukhatme VP, Karumanchi SA. Angiogenic imbalance in the pathophysiology of preeclampsia: newer insights. *Semin Nephrol.* 2004;24:548–556.)

Derangements in other angiogenic molecules have also been observed. Levels of endostatin, another antiangiogenic factor, are elevated in preeclampsia.[171] Expression of $VEGF_{165b}$, an antiangiogenic isoform of VEGF, are reduced in the first trimester in women who later develop preeclampsia.[172] Circulating levels of soluble endoglin (sEng), a truncated form of the transforming growth factor-β (TGF-β) receptor, are elevated in preeclampsia. sEng amplifies the vascular damage mediated by sFlt1 in pregnant rats, inducing a severe preeclampsia-like syndrome with features of the HELLP syndrome.[173] Maternal serum levels of sEng rise prior to preeclampsia onset,[174–176] in a pattern similar to sFlt1.[177] As TGF-β regulates podocyte VEGF-A expression,[178] sEng may result in impaired local VEGF signaling in the glomerulus. Notably, individuals who were born preterm have elevations in sEng, as well as sFlt1, as compared with individuals who were born term to uncomplicated pregnancies (intergenerational programming of preeclampsia risk is further discussed in Chapter 21).[179] It was recently reported that semaphorin 3B, a novel trophoblastic-secreted antiangiogenic protein, was also upregulated in preeclamptic placentas and that semaphorin 3B inhibits trophoblast migration and invasion by downregulating VEGF signaling.[180] The precise role of these novel antiangiogenic proteins and their relationship with sFlt1 in the systemic vasculature continues to be explored.

There is circumstantial evidence suggesting that interference with VEGF signaling may lead to preeclampsia.[160] VEGF appears to be important in the stabilization of endothelial

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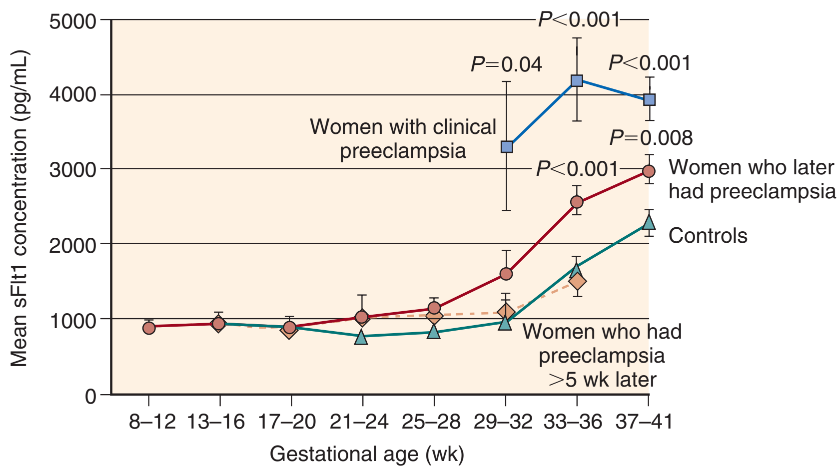

Fig. 48.11 Concentrations of soluble fms-like tyrosine kinase-1 (sFlt1) in preeclampsia and normal pregnancy. Shown are the mean serum sFlt1 concentrations (± standard error of mean) before and after onset of clinical preeclampsia according to the gestational age of the fetus. The *P* values given are for comparisons, after logarithmic transformation, with specimens from controls obtained during the same gestational-age interval. All specimens were obtained before labor and delivery. (From Levine RJ, Maynard SE, Qian C, et al. Circulating angiogenic factors and the risk of preeclampsia. *N Engl J Med.* 2004;350:672–683.)

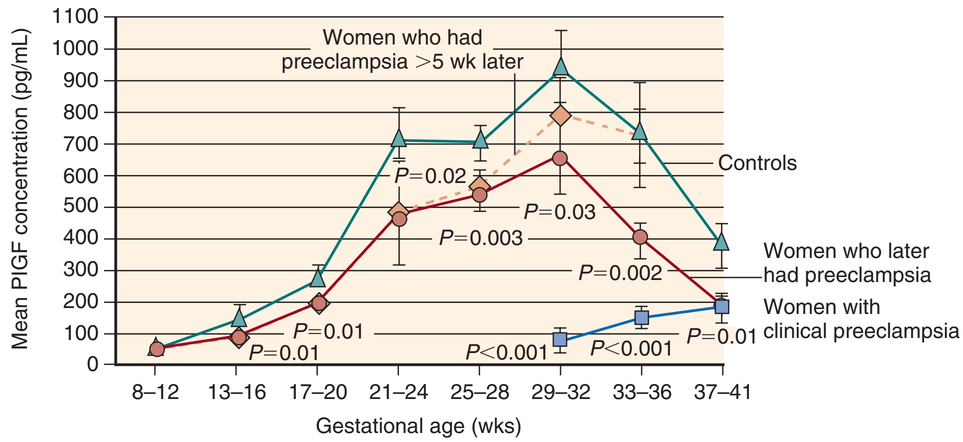

Fig. 48.12 Concentrations of placental growth factor (PlGF) in preeclampsia and normal pregnancy. Shown are the mean serum PlGF concentrations (± standard error of the mean) before and after onset of clinical preeclampsia according to the gestational age of the fetus. The *P* values given are for comparisons, after logarithmic transformation, with specimens from controls obtained during the same gestational-age interval. All specimens were obtained before labor and delivery. (From Levine RJ, Maynard SE, Qian C, et al. Circulating angiogenic factors and the risk of preeclampsia. *N Engl J Med.* 2004;350:672–683.)

cells in mature blood vessels. VEGF is particularly important in the health of the fenestrated and sinusoidal endothelium found in the renal glomerulus, brain, and liver[181,182] – organs disproportionately affected in preeclampsia. VEGF is highly expressed by glomerular podocytes, and VEGF receptors are present on glomerular endothelial cells.[183] In experimental glomerulonephritis, VEGF is necessary for glomerular capillary repair.[184] In a podocyte-specific VEGF knockout mouse, heterozygosity for VEGF-A resulted in renal disease characterized by proteinuria and glomerular endotheliosis.[185] In antiangiogenesis cancer trials, VEGF antagonists produce proteinuria and hypertension in human subjects.[186–189] This evidence suggests that VEGF deficiency induced by excess sFlt1 has the capacity to produce the characteristic renal lesion of preeclampsia.

Although the glomerular capillary endothelial cell appears to be the primary glomerular target in preeclampsia, the podocyte is clearly affected in severe disease as evidenced by podocyturia during clinical disease and even before overt proteinuria.[140,190] However, podocyturia is also noted in other proteinuric disorders and is not specific for preeclampsia. Renal autopsy examination from women who died with preeclampsia demonstrates markedly reduced podocyte expression of nephrin,[191] and serum from preeclamptic women reduces nephrin expression by cultured podocytes.[192] Using a mouse model of preeclampsia induced by administration of anti-VEGF antibodies, it was noted that nephrin expression is reduced in podocytes.[193] The effect of sFlt1 on podocyte nephrin expression appears to be via increased endothelin-1 release by endothelial cells,[192] again implicating the glomerular endothelial cell as the primary site of injury in preeclampsia.

Alterations in circulating sFlt1 have been noted in certain preeclampsia risk groups. Higher sFlt1 levels have been noted in first versus second pregnancies,[194] in twin versus singleton

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pregnancies,[195,196] in women with prior preeclampsia,[197] and in women carrying fetuses affected by trisomy 13,[198,199] potentially accounting for the increased preeclampsia risk in these groups. In the case of twin pregnancies, the increased sFlt1 production appears to be due to increased placental mass, rather than placental ischemia.[196] Conversely, lower levels of sFlt1 in pregnant smokers[177,200] may explain the protective effect of smoking in preeclampsia.[201,202]

Angiogenic factors are likely to be important in the regulation of placental vasculogenesis. VEGF ligands and receptors are highly expressed by placental tissue in the first trimester.[203] sFlt1 decreases cytotrophoblast invasiveness in vitro.[123] Circulating sFlt1 levels are relatively low early in pregnancy, and begin to rise in the third trimester. This may reflect a physiologic antiangiogenic shift in the placental milieu toward the end of pregnancy, corresponding to completion of the vasculogenic phase of placental growth. It is intuitive to hypothesize that placental vascular development might be regulated by a local balance between proangiogenic and antiangiogenic factors, and that excess antiangiogenic sFlt1 in early gestation could contribute to inadequate cytotrophoblast invasion in preeclampsia. By the third trimester, excess placental sFlt1 is detectable in the maternal circulation, producing end-organ effects. In this case, placental ischemia may not be causative, but rather the earliest organ affected by this derangement of angiogenic balance.

The pathways regulating placental angiogenic factor expression, and the reasons for their dysregulation in preeclampsia, are yet unknown (see Fig. 48.8). Pieces of the puzzle may be starting to emerge, however. Animal models of preeclampsia based on induction of uteroplacental ischemia are characterized by increased endogenous sFlt1[116,204] and sEng.[205] Agonistic AT_1 autoantibodies produce a preeclampsia-like syndrome in mice (see later), in association with increased circulating sFlt1 levels.[206] Elaboration of the upstream pathways involved in placental angiogenic factor expression remains an area of intense research. Heme oxygenase 1 and its downstream metabolite carbon monoxide acts as a vascular protective factor by inhibiting the production of sFlt1.[207] Animal studies suggest that cystathionine γ-lyase (that regulates hydrogen sulfide) may also contribute to preeclampsia by disrupting placental angiogenesis.[208] However, human studies demonstrating alterations in heme oxygenase or cystathionine γ-lyase expression prior to alterations in antiangiogenic factor are still lacking.

In addition to angiogenic alterations, women who develop preeclampsia also have evidence of insulin resistance.[209] Moreover, women with pregestational or gestational diabetes mellitus have an increased risk for developing preeclampsia.[53] In accordance with this finding, in vitro models suggest that insulin signaling and angiogenesis are intimately related at a molecular level,[210] and recent epidemiological data find that evidence of altered angiogenesis and excess insulin resistance may be additive insults that lead to preeclampsia.[211] Furthermore, altered levels of biomarkers linked with angiogenesis and insulin resistance persist in the postpartum state,[212] possibly explaining the long-term cardiovascular risk in these women.

ANGIOTENSIN-1 RECEPTOR AUTOANTIBODIES

In preeclampsia, plasma renin levels are suppressed relative to normal pregnancy as a secondary response to systemic vasoconstriction and hypertension. As noted in the prior section, preeclampsia is characterized by increased vascular responsiveness to angiotensin II and other vasoconstrictive agents. Wallukat et al[213] identified agonistic angiotensin-1 (AT_1) receptor autoantibodies in women with preeclampsia. They hypothesized that these antibodies, which activate the AT_1 receptor, may account for the increased angiotensin II sensitivity of preeclampsia. The same investigators later showed that these AT_1 autoantibodies, like angiotensin II itself, stimulate endothelial cells to produce tissue factor, an early marker of endothelial dysfunction. Xia et al[214] found that AT_1 autoantibodies decreased invasiveness of immortalized human trophoblasts in an in vitro invasion assay, suggesting that these autoantibodies might contribute to defective placental pseudovasculogenesis as well. The same group showed that AT_1 autoantibodies isolated from the serum of women with preeclampsia produce a preeclampsia-like syndrome in pregnant rats, with increased endogenous sFlt1 production,[206] suggesting that AT_1 autoantibodies and sFlt1 may be part of the same pathophysiologic pathway leading to preeclampsia. Increased endogenous sFlt1 and the phenotype of proteinuria and glomerular endotheliosis were seen in the pregnant dams but not in nonpregnant animals, providing further evidence that placental sFlt1 specifically mediates the proteinuria secondary to AT_1 autoantibodies.

AT_1 autoantibodies are not limited to pregnancy; they also appear to be increased in malignant renovascular hypertension in nonpregnancy.[215] In addition, these antibodies have been identified in women with abnormal second-trimester uterine artery Doppler studies who did not develop preeclampsia, suggesting that this antibody may be a nonspecific response to placental hypoperfusion.[216]

The angiotensinogen T235 polymorphism, a common molecular variant associated with essential hypertension and microvascular disease, has been associated with preeclampsia in several studies across different populations, though with significant ethnic heterogeneity.[217] Functional implications of this polymorphism remain unclear. Work by Abdalla and colleagues[218] have suggested that heterodimerization of AT_1 receptors with bradykinin 2 receptors may contribute to angiotensin II hypersensitivity in preeclampsia. This work remains to be validated in other studies.

SCREENING

Although there is not yet any definitive therapeutic or preventive strategy for preeclampsia, clinical experience suggests that early detection, monitoring, and supportive care is beneficial to the patient and the fetus. For example, lack of adequate antenatal care is strongly associated with poor outcomes, including eclampsia and fetal death.[219] Risk assessment early in pregnancy is important to identify those who require close monitoring after 20 weeks. Women with first pregnancies or other preeclampsia risk factors (Table 48.2) should be assessed frequently after 20 weeks' gestation for the development of hypertension, proteinuria, headache, visual disturbances, or epigastric pain.

Higher BP in the first or second trimesters, even in the absence of overt hypertension, is associated with elevated risk for preeclampsia in healthy nulliparous women.[220] Unfortunately, these small elevations in midtrimester BP are

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subtle and the positive predictive value as a screening test is low (especially given the relatively low prevalence), limiting routine clinical utility.

Presumably as a result of failed placental vascular remodeling, preeclampsia is associated with increased placental vascular resistance and uterine artery waveform abnormalities in the second trimester, as measured by uterine artery Doppler ultrasound.[221] Dozens of studies have investigated the use of uterine artery Doppler for prediction of preeclampsia. Test performance varies widely among studies based on differences in populations studied, the gestational age at the time of measurement, the definition of an abnormal result, and the severity and timing of preeclampsia detected: sensitivities and specificities range from 65% to 85%. Even meta-analyses have differed in their conclusions, with some reporting limited diagnostic accuracy in predicting preeclampsia,[221,222] while others suggest that it is accurate enough to be recommended for preeclampsia screening in routine clinical practice.[223] Thus there is wide regional variability in the use of uterine artery Doppler for routine screening, and it remains uncommon in the United States. Recent data suggest that there may be promise in combining uterine artery Doppler with serum biomarkers for preeclampsia.[224,225]

Of dozens of putative serum markers for preeclampsia, only a handful have been shown to be elevated prior to the onset of clinical disease, and none have yet proven to be an effective and useful screening test for preeclampsia. Placental protein 13 (PP-13) and angiogenic biomarkers (sFlt1, PlGF, and sEng) hold promise for screening and/or early diagnosis of preeclampsia.

PP-13 is thought to be involved in normal placentation and maternal vascular remodeling. Low levels of PP-13 could identify women at high risk for preeclampsia as early as the first trimester,[226,227] although it appears to be a robust biomarker only for early onset disease and may be less useful for preeclampsia close to term.[228] Polymorphisms in *LGALS13*, the gene encoding PP-13, have been detected in cases of preeclampsia.[229] This polymorphism may result in production of a shorter splice variant of PP-13, which is not detected by conventional assays, contributing to low circulating levels and decreased local activity of PP-13.

Alterations in circulating levels of the angiogenic factors sFlt1 and sEng occur weeks prior to the onset of preeclampsia and may be useful for screening and/or diagnosis.[230,231] Significant elevations in maternal sFlt1 and sEng are observed from midgestation onward,[162,174–177,232,233] and appear to rise 5–8 weeks prior to preeclampsia onset (see Fig. 48.11).[164] Maternal sFlt1 levels are particularly elevated in severe preeclampsia, early onset preeclampsia, and preeclampsia complicated by a small-for-gestation infant.[164,234] Serum levels of PlGF are lower in women who go on to develop preeclampsia from the first[235] or early second[164,236–238] trimester (see Fig. 48.12). Since PlGF passes into the urine, low urinary PlGF has been identified as a potential marker for preeclampsia. Urinary levels of PlGF are significantly lower in women who develop preeclampsia from the late second trimester,[239] and may prove to be useful in screening and diagnosis of preeclampsia, especially in early onset and severe preeclampsia. Prospective studies are ongoing to evaluate the clinical utility of these biomarkers for preeclampsia screening and risk assessment.

Clinical Relevance

Preeclampsia is characterized by marked alterations in angiogenic factors. Measurement of maternal serum sFlt1, placental growth factor (PlGF), and the sFlt1-to-PlGF ratio may inform diagnosis and risk stratification in women with suspected preeclampsia.

Recent studies have suggested that circulating angiogenic factors in plasma or serum can be used to differentiate preeclampsia from other diseases that mimic preeclampsia such as chronic hypertension, gestational hypertension, lupus nephritis, and CKD.[240–244] Zeisler et al[245] demonstrated in a prospective multicenter clinical trial that serum sFlt/PlGF can be used to rule out preeclampsia among patients with suspected disease with negative predictive value >99%. Several groups have also demonstrated a role for angiogenic biomarkers in the prediction of preeclampsia-related adverse outcomes among women evaluated for suspected preeclampsia.[246–250] Measurements of circulating angiogenic factors (sFlt1, PlGF, and sEng) robustly predicted adverse maternal and perinatal outcomes in women presenting with signs or symptoms of preeclampsia and these biomarkers outperformed the standard battery of clinical diagnostic measures including BP, proteinuria, uric acid, and other laboratory assays. Importantly, sFlt1 and/or PlGF levels at presentation were strongly associated with the remaining duration of pregnancy.[246,249] By providing more accurate identification of women at high risk for adverse pregnancy outcomes, use of angiogenic biomarkers may reduce costs and unnecessary resource use in women with possible preeclampsia.[251] For example, women with mild-to-moderate preeclampsia, in whom biomarker testing indicates low risk of imminent delivery or severe complications, may be able to be safely discharged and observed in an outpatient setting, rather than remain under indefinite inpatient observation until delivery.

PREVENTION OF PREECLAMPSIA

ANTIPLATELET AGENTS

Aspirin for the prevention of preeclampsia has been evaluated in dozens of trials, both in high-risk and in healthy nulliparous women. The most recent meta-analyses suggest that aspirin reduces the risk of preeclampsia by at least 10%.[252] Some analyses suggest a risk reduction closer to 24% when aspirin is initiated prior to 16 weeks' gestation, and using a dosage of at least 100 mg daily (Fig. 48.13).[253] The absolute risk reduction is greatest among women at high baseline preeclampsia risk; thus current guidelines recommend treatment with acetylsalicylic acid (aspirin) in women with at least one major, or two moderate, preeclampsia risk factors.[254] Based on consistent safety and efficacy in both high- and low-risk populations, some experts recommend universal aspirin for all pregnant women, regardless of baseline preeclampsia risk.[255] Recent data on long-term follow-up of children exposed to low-dose aspirin in utero demonstrate a modest risk of childhood asthma.[256] This potential risk should be considered before widespread use of aspirin for those at low risk of preeclampsia is universally advocated. Rolnik et al[257] recently published results with an impressive 62% reduction in the risk of preterm

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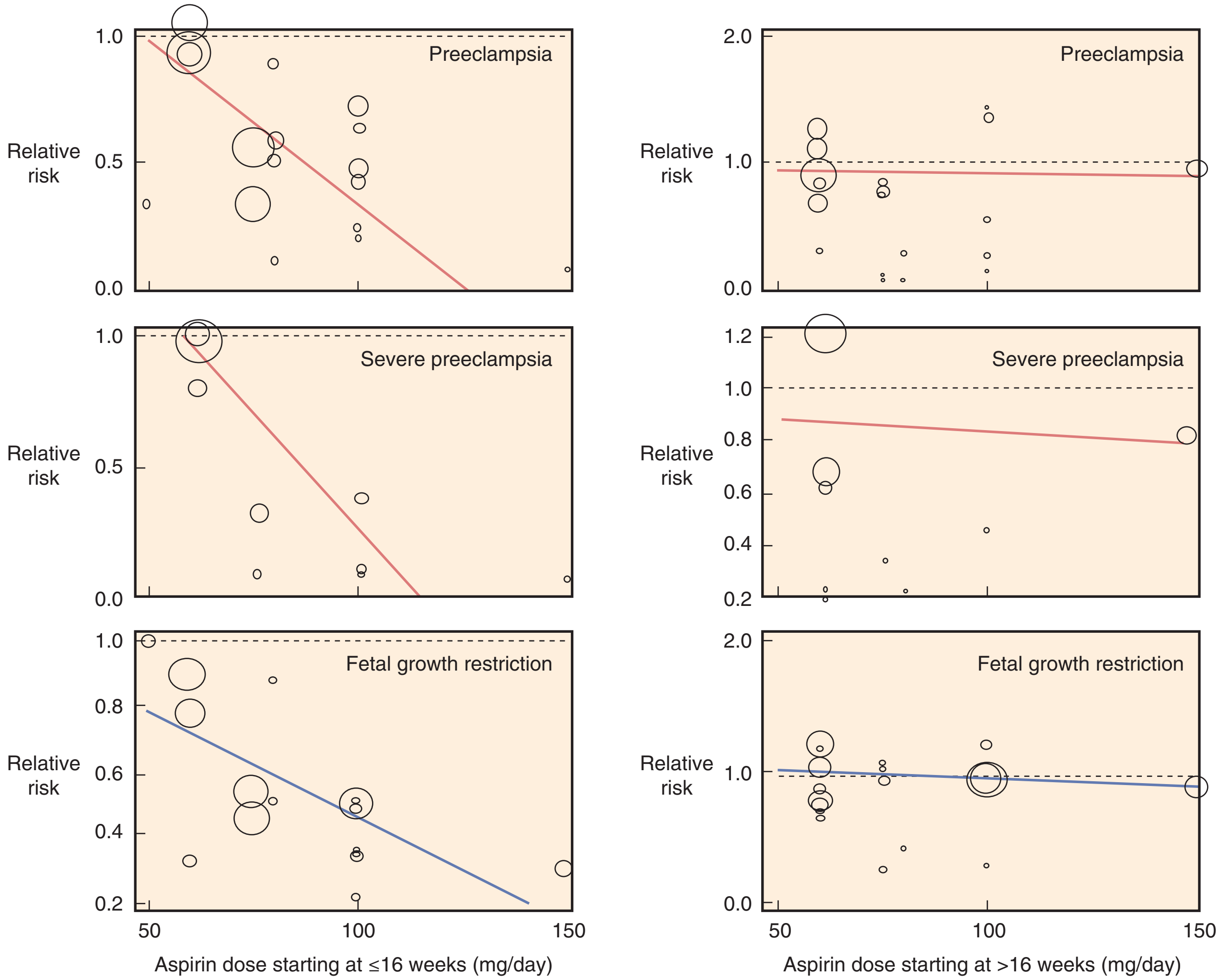

Fig. 48.13 The effect of aspirin on preeclampsia, severe preeclampsia, and fetal growth restriction. Bubble plots with fitted metaregression line showing the relationship between aspirin dosage and relative risks for each adverse pregnancy outcome, according to gestational age at initiation of aspirin therapy. Data are based on meta-analysis of 45 randomized controlled trials, including 20,909 pregnant women. *RR,* Relative risk. (From Roberge S, Nicolaides K, Demers S, et al. The role of aspirin dose on the prevention of preeclampsia and fetal growth restriction: systematic review and meta-analysis. *Am J Obstet Gynecol*. 2017;216:110–120.e6.)

preeclampsia with utilization of aspirin (150 mg) in high-risk women identified by a previously studied algorithmic score combining mean arterial pressure, uterine-artery pulsatility index, biochemical markers – PlGF and pregnancy-associated plasma protein-A. This new trial provides early evidence that a select group of women with a higher predicted risk of preterm preeclampsia using biophysical algorithms may greatly benefit from receiving aspirin prophylaxis.

CALCIUM FOR THE PREVENTION OF PREECLAMPSIA

Low baseline dietary calcium intake is associated with increased preeclampsia risk.[258] Several studies have examined the effectiveness of calcium supplementation to prevent preeclampsia. Among low-risk primiparous North American women, calcium supplementation did not reduce incidence of preeclampsia.[61] However, a metaanalysis of 13 trials, including 15,730 women from various parts of the world, reported a significant reduction in preeclampsia risk with high-dose (>1 g/day) calcium supplementation [RR 0.45, 95% confidence interval (CI) 0.31–0.65], with the greatest effect among women with low baseline calcium intake and at high preeclampsia risk.[259] This was tested directly in a large randomized, placebo-controlled trial of calcium supplementation in >8000 women with low baseline calcium intake (<600 mg/day).[260] Although there was no difference in the incidence of preeclampsia, the calcium group had a lower rate of eclampsia, gestational hypertension, preeclampsia complications, and neonatal mortality. Thus calcium supplementation may be useful, especially in women with low baseline calcium intake.

ANTIOXIDANTS AND NUTRITIONAL INTERVENTIONS

Based on the hypothesis that oxidative stress may contribute to pathogenesis, it has been suggested that antioxidants may prevent preeclampsia.[261] However, four large randomized, controlled trials have failed to show a benefit of vitamin C and vitamin E supplementation for the prevention of preeclampsia in various populations,[262–266] even in women at high preeclampsia risk recruited from communities at risk for poor nutritional status.[262] Vitamin D deficiency has been

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associated with preeclampsia in some studies.[267–269] However, a randomized, controlled trial failed to demonstrate a reduction in preeclampsia with vitamin D supplementation.[270] More work is needed to evaluate whether treatment of vitamin D insufficiency reduces preeclampsia risk.

Nutritional interventions have generally not been effective in decreasing preeclampsia risk. Protein and calorie restriction for obese pregnant women shows no reduction in the risk of preeclampsia or gestational hypertension, and may increase risk for IUGR, so should be avoided.[271] However, obese women with lower gestational weight gain (<15 kg) have a reduced incidence of preeclampsia.[272,273] Women who have had bariatric surgery for severe obesity have a reduced incidence of preeclampsia compared with obese controls in some,[274] but not in all,[275] studies.

LOW-MOLECULAR-WEIGHT HEPARIN

Low-molecular-weight heparin (LMWH) is commonly used to prevent recurrent pregnancy loss in women with antiphospholipid syndrome and inherited thrombophilias, such as factor V Leiden mutation. Several studies have evaluated the potential benefit of LMWH for the prevention of preeclampsia and other placental-mediated pregnancy complications in women with a history of these complications in prior pregnancies. A meta-analysis summarized six of these randomized, controlled trials.[276] The analysis suggested a significant reduction in the composite outcome of preeclampsia, birth of a small-for-gestational-age newborn (<10th percentile), placental abruption, or pregnancy loss >20 weeks in women treated with prophylactic LMWH (18.7% of treated women vs. 42.9% of controls). However, a subsequent multicenter open-label randomized controlled trial failed to show a reduction in the incidence of preeclampsia or IUGR among high-risk women treated with LMWH.[277] More studies are needed before LMWH can be recommended for preeclampsia prevention in high-risk women.

MANAGEMENT AND TREATMENT OF PREECLAMPSIA

TIMING OF DELIVERY

The timing of delivery in severe preeclampsia is controversial. In women presenting prior to 24 weeks' gestation, perinatal and neonatal mortality are extremely high (>80%) even with attempts to postpone delivery, and maternal complications are common.[278] For this reason, termination is usually recommended in women with severe preeclampsia presenting prior to 24 weeks. By contrast, in women presenting after 37 weeks' gestation, the neonatal benefit of prolonging pregnancy is minimal, and immediate delivery is almost always indicated to avoid potential maternal and fetal complications. For women presenting between 24 and 37 weeks' gestation, the potential neonatal benefit of continuing pregnancy must be balanced against the possibility of maternal morbidity and mortality with delaying delivery. In general, the presence of nonreassuring fetal testing; suspected abruptio placentae; thrombocytopenia; worsening liver and/or kidney function; and symptoms such as unremitting headache, visual changes, nausea, vomiting, or epigastric pain are generally considered indications for expedient delivery.

Two small, randomized controlled trials demonstrated that in women presenting with severe preeclampsia between 28 and 32 weeks' gestation, expectant management (with delivery postponed 1–2 weeks after presentation) resulted in decreased neonatal complications and decreased neonatal intensive care unit stay, with no significant increase in maternal complications.[279,280] Subsequent observational studies have confirmed that delivery can be safely and effectively postponed in women with severe preeclampsia with careful and intensive fetal and maternal monitoring.[278] Similarly, in women with nonsevere preeclampsia presenting between 34 and 37 weeks' gestation, a strategy of expectant monitoring (aimed at prolonging pregnancy until 37 weeks' gestation) led to a reduction in neonatal respiratory distress syndrome, without apparent adverse maternal effects.[281] Expectant management also appears to decrease subsequent respiratory disorders in childhood.[282]

There are no randomized controlled trials to evaluate optimal mode of delivery in severe preeclampsia. Retrospective studies suggest that maternal and neonatal outcomes are similar among women undergoing induction of labor and those undergoing cesarean section.[283]

BLOOD PRESSURE MANAGEMENT

Management of BP in preeclampsia is substantially different from that in the nonpregnant population. Rather than seeking to minimize long-term cerebrovascular and cardiovascular complications, the goal of care is to maximize the likelihood of successful delivery of a healthy infant, while minimizing the chance of acute complications in the mother. Acute aggressive lowering of BP can lead to fetal distress or demise, especially if placental perfusion is already compromised. Because of this, antihypertensive therapy for preeclampsia is usually withheld unless the BP rises above 150–160 mm Hg systolic or 100–110 mm Hg diastolic, above which the risk of cerebral hemorrhage becomes significant.[68] In the next section we review details regarding the use of specific antihypertensive agents for BP management in pregnancy.

MAGNESIUM AND SEIZURE PROPHYLAXIS

Magnesium has been widely used for the management and prevention of eclampsia for decades. Prior to the mid-1990s, evidence for its use was largely derived from clinical experience and from small uncontrolled studies. Over the last 10 years, magnesium has been proven to be superior to other agents in the prevention and treatment of seizures in preeclampsia, although not for the prevention of preeclampsia per se. In 1995, two randomized controlled trials in an international and a US population showed that magnesium sulfate was superior to diazepam and phenytoin in reducing risk of seizures in women with preeclampsia/eclampsia.[284,285] Subsequently, the Magpie study confirmed this benefit with a randomized, controlled trial of magnesium versus placebo for seizure prevention in >10,000 women with preeclampsia from 33 countries, finding that magnesium decreased the incidence of eclamptic seizure by 50% (0.8% vs. 1.9%).[88] A global public health effort over the past 10–15 years has led to improved access to magnesium in developing countries, where use of magnesium in preeclampsia/eclampsia has now reached 75% to 90%.[286]

Magnesium is now categorized by the US Food and Drug Administration with a warning label, due to adverse fetal bone effects with long-term (>5–7 days) use as a tocolytic in

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preterm labor (http://www.fda.gov/Drugs/DrugSafety/ucm353333.htm). Despite this labeling change, short-term (<48 hours) intravenous magnesium is still recommended by the American College of Obstetrics and Gynecology for the prevention and treatment of seizures in women with preeclampsia and eclampsia.[287] Magnesium is generally given intravenously as a bolus, followed by a continuous infusion. In the therapeutic range (5 to 9 mg/dL), magnesium sulfate slows neuromuscular conduction and depresses central nervous system irritability. Women receiving continuous infusions of magnesium should be monitored carefully for signs of toxicity, including loss of deep tendon reflexes, flushing, somnolence, muscle weakness, and decreased respiratory rate. Such monitoring is especially important in women with impaired renal function who have impaired urinary magnesium excretion.

MANAGEMENT OF THE HELLP SYNDROME

The clinical course of the HELLP syndrome usually involves inexorable and often sudden and unpredictable deterioration. Given the high incidence of maternal complications, some authors recommend immediate delivery in all cases of confirmed HELLP. Among women in the 24–34 weeks' gestational window whose clinical status appears relatively stable and with reassuring fetal status, expectant management is often a viable alternative. For many years intravenous steroids have been suggested as an adjunct to usual management, based on retrospective and uncontrolled studies. A recent randomized controlled trial showed no benefit to high-dose dexamethasone treatment in HELLP syndrome.[288] Post hoc analysis suggested that the subgroup with severe preeclampsia (platelet count < 50,000) may have a shorter average platelet count recovery and shorter hospitalization with steroids, so further studies are required to evaluate the benefit in this population.

Based on pathophysiologic similarities to TTP, there are reports of using plasmapheresis in the management of the HELLP syndrome. Data for the antepartum period are limited to a few cases series with mixed results and no clear benefit.[289] Potential drawbacks include fetal compromise due to diminishment of already compromised placental blood flow.

NOVEL THERAPIES FOR PREECLAMPSIA

Recent advances in our understanding of the pathophysiology of preeclampsia have revealed new potential therapeutic targets. Interfering with the production or signaling of sFlt1 may ameliorate the endothelial dysfunction of preeclampsia, allowing delivery to be more safely postponed. In a pilot study limited to three severe early preeclamptics (24–32 weeks' gestation), Thadhani et al[290] depleted sFlt1 levels using dextran sulfate apheresis and prolonged pregnancy by 2–4 weeks. Importantly, this therapy promoted fetal growth with no adverse effects to the fetus or the mother. More recently, Thadhani et al[291] demonstrated that therapeutic apheresis using a plasma-specific dextran sulfate column produced a 18% reduction in sFlt1, a 44% reduction in P:C ratio, and a prolongation of pregnancy as compared with untreated contemporaneous preeclampsia subjects. If confirmed, this approach could lead to targeted therapy for patients with preterm preeclampsia that present with an abnormal angiogenic profile.[290] Other modalities for targeting the angiogenic imbalance are administration of agents that neutralize sFlt1 such as PlGF and VEGF, or small molecules that block sFlt1 production, which are currently being evaluated in preclinical settings.[160,292–295]

Hydroxymethylglutaryl-coenzyme A reductase inhibitors, or statins, have been proposed as potential therapeutic agents for preeclampsia through their effects on promoting heme oxygenase activity and improving angiogenic imbalance in animal models of preeclampsia.[207,296] These agents were once considered contraindicated in pregnancy due to potential teratogenic effects, but recent data suggest that they are probably safe.[297] Early pilot data suggest that statins may be effective clinically both for the treatment[298] and for the prevention[299] of preeclampsia. Randomized, controlled trials to assess the safety and efficacy of pravastatin for the prevention and treatment of preeclampsia are ongoing.[300]

CHRONIC HYPERTENSION AND GESTATIONAL HYPERTENSION

The diagnosis of chronic hypertension in pregnancy is usually based on a documented history of hypertension prior to pregnancy or a BP >140/90 prior to 20 weeks' gestation. Gestational hypertension, by contrast, is usually first noted after 20 weeks' gestation and, by definition, resolves after delivery. These diagnoses, based on the timing of the first recorded BP elevation, can be subject to pitfalls, however. The physiologic dip in BP in the second trimester, which nadirs at about 20 weeks' gestation (see Fig. 48.1), occurs in women with chronic hypertension and can mask the presence of underlying chronic hypertension early in pregnancy. In such cases, a woman with chronic hypertension may be inappropriately labeled as having gestational hypertension when the BP rises in the third trimester. The diagnosis of chronic hypertension is established when hypertension fails to resolve postpartum. By contrast, preeclampsia can occasionally present prior to 20 weeks' gestation; hence preeclampsia should always be considered in women presenting with new hypertension and proteinuria close to midgestation.

The incidence of chronic hypertension in pregnancy is increasing in the United States, from 1.01% in 1995–1996 to 1.76% in 2007–2008.[301] Chronic hypertension is more common with advanced maternal age, obesity, and black race.[302] Pregnant women with chronic hypertension have an increased risk of preeclampsia (21%–25%), premature delivery (33%–35%), IUGR (10%–15%), placental abruption (1%–3%), and perinatal mortality (4.5%).[303–305] However, most adverse outcomes occur in women with severe hypertension (DBP > 110 mm Hg) and those with preexisting cardiovascular or renal disease. Women with mild, uncomplicated chronic hypertension usually have obstetric outcomes comparable with the general obstetric population.[302] Both the duration and the severity of hypertension are correlated with perinatal morbidity and preeclampsia risk.[306,307] The presence of baseline proteinuria increases the risk of preterm delivery and IUGR, but not preeclampsia per se.[304]

The diagnosis of preeclampsia superimposed on chronic hypertension can be difficult. In the absence of underlying kidney disease, the new onset of proteinuria (>300 mg/day), usually with worsening hypertension, is the most reliable sign of superimposed preeclampsia.[68] When proteinuria is

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present at baseline, preeclampsia is likely when a sudden increase in BP is observed in a woman whose BP was previously well-controlled. The presence of other signs and symptoms of preeclampsia such as headache, visual changes, epigastric pain, pulmonary edema, and laboratory derangements (e.g., thrombocytopenia, new or worsening renal insufficiency, and elevated liver enzymes) also should prompt consideration of preeclampsia, and when present, are an indication of preeclampsia severity.[68]

SECONDARY HYPERTENSION IN PREGNANCY

Secondary causes of hypertension are present in at least 10% of women with chronic hypertension in pregnancy.[301] Women with secondary forms of hypertension have pregnancy complication rates even higher than women with primary chronic hypertension, and failing to recognize these diagnoses can lead to significant maternal morbidity and mortality. Hence prepregnancy evaluation of women with chronic hypertension should include consideration of secondary causes of hypertension, including renal artery stenosis, primary hyperaldosteronism, obstructive sleep apnea (OSA), and pheochromocytoma. Unfortunately, diagnosis of these conditions during pregnancy is hampered by the fact that screening tests (e.g., the plasma aldosterone-to-renin ratio) have not been validated in pregnancy, and imaging involving radiation exposure (e.g., CT scanning, angiography, and fluoroscopy) is relatively contraindicated.

Renal artery stenosis due to fibromuscular dysplasia or atherosclerotic vascular disease occasionally presents in pregnancy, and should be suspected when hypertension is severe and resistant to medical therapy. Diagnosis with MR angiography followed by successful angioplasty and stent placement in the second and third trimesters of pregnancy has been described.[308]

Although rare, pheochromocytoma can be devastating if it first presents during pregnancy. This syndrome occasionally is unmasked during labor and delivery, when fatal hypertensive crisis can be triggered by vaginal delivery, uterine contractions, and anesthesia.[309] Maternal and neonatal outcomes are much better when the diagnosis is made antepartum, with attentive and aggressive medical management.[310] Surgical intervention is typically postponed until after delivery whenever possible.

Hypertension and hypokalemia from primary hyperaldosteronism might be expected to improve during pregnancy, as progesterone antagonizes the effect of aldosterone on the renal tubule. However, such remission is not universal and many women with primary hyperaldosteronism have a pregnancy-induced exacerbation of hypertension.[311] In the case of a functional adrenal adenoma, there are little data to favor either immediate surgical adrenalectomy or medical management until after delivery, though case reports have suggested success with both approaches. Although the use of spironolactone has been reported during pregnancy, there is a theoretical risk of inadequate virilization of male fetuses due to its antiandrogenic effects. Eplerenone appears to be a safe alternative in cases where conventional antihypertensive agents and potassium supplementation are inadequate.[312]

OSA has emerged as an important secondary cause of hypertension in pregnancy. In one study, 40% of women with hypertension in pregnancy had evidence of OSA by polysomnography.[313] Risk factors for OSA include snoring and obesity.[314] Screening for OSA should be considered in high-risk women.

A rare cause of early onset hypertension is due to a mutation in the mineralocorticoid receptor. This results in inappropriate receptor activation by progesterone, and affected women develop a marked exacerbation of hypertension and hypokalemia in pregnancy, but without proteinuria or other features of preeclampsia.[315]

APPROACH TO MANAGEMENT OF CHRONIC HYPERTENSION IN PREGNANCY

In women with chronic hypertension who are planning pregnancy, BP should be controlled prior to conception using agents that are safe in pregnancy. When pregnancy is unplanned, inappropriate antihypertensive medications should be stopped as soon as possible after conception. Women should be counseled regarding the risks of adverse pregnancy outcomes, including preeclampsia, preterm birth, and IUGR. Women with chronic hypertension in pregnancy should be followed closely during pregnancy, with appropriate adjustments in antihypertensive therapy, and monitoring for superimposed preeclampsia.

GOALS OF THERAPY

When hypertension is severe (SBP ≥ 160 or DBP ≥ 105 mm Hg), antihypertensive therapy is clearly indicated for the prevention of stroke and cardiovascular complications.[68] However, there is little evidence to guide management of mild-to-moderate hypertension in pregnancy. Dozens of small ($n < 300$) clinical trials have evaluated the impact of antihypertensive therapy versus no treatment in such women, and these have been evaluated in meta-analyses.[316–319] Although antihypertensive therapy lowered the risk of developing severe hypertension, there was no beneficial effect on the development of preeclampsia, neonatal death, preterm birth, small for gestational age babies, or other adverse outcomes. Some studies suggested that aggressive treatment of mild-to-moderate hypertension in pregnancy may impair fetal growth, presumably as a result of decreased uteroplacental perfusion.[320] For this reason, the 2013 American College of Obstetrics and Gynecology Task Force on Hypertension in Pregnancy recommends against antihypertensive medication use in pregnant women with chronic hypertension and BP < 160/105 mm Hg in the absence of evidence of end-organ damage.[68]

Clinical Relevance

The CHIPS (Control of Hypertension in Pregnancy Study) trial showed that treatment of hypertension in pregnancy to a “tight” blood pressure target [diastolic blood pressure (DBP) 85 mm Hg], as compared with a “less-tight” target (DBP 100 mm Hg), was safe, with no significant differences in the rate of pregnancy loss or need for high-level neonatal care. Women treated to the tight blood pressure target had a significantly lower rate of maternal complications, including severe hypertension, thrombocytopenia, and transaminitis with symptoms.

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Table 48.5 Outcomes in Women With Hypertension in Pregnancies Randomized to Tight Versus Less-Tight Blood Pressure Control

Outcome	Less-Tight Control (DBP 100 mm Hg)	Tight Control (DBP 85 mm Hg)	Adjusted Odds Ratio (95% CI)
Neonatal and Pregnancy Complications			
Pregnancy loss, *n* (%)	15 (3.0)	13 (2.7)	1.14 (0.53–2.45)
High-level neonatal care for >48 h *n* (%)	141 (29.4)	139 (29.0)	1.00 (0.75–1.33)
Gestational age at delivery (wk)	36.8 ± 3.4	37.2 ± 3.1	
Small-for-gestational-age newborns (birth weight < 10th percentile), *n* (%)	79 (16.1)	96 (19.7)	0.78 (0.56–1.08)
Placental abruption	11 (2.2)	11 (2.3)	0.94 (0.40–2.21)
Maternal Complications			
Serious maternal complications, *n* (%)	18 (3.7)	10 (2.0)	1.74 (0.79–3.84)
Severe hypertension	200 (40.6)	134 (27.5)	1.80 (1.34–2.38)
Preeclampsia	241 (48.9)	223 (45.7)	1.14 (0.88–1.47)
Platelet count < 100 × 10^9/L	21 (4.3)	8 (1.6)	2.63 (1.15–6.05)
Elevated AST or ALT with symptoms	21 (4.2)	9 (1.8)	2.33 (1.05–5.16)
HELLP syndrome	9 (1.8)	2 (0.4)	4.35 (0.93–20.35)

ALT, Alanine aminotransferase; *AST,* aspartate aminotransferase; *CI,* confidence interval; *DBP,* diastolic blood pressure; *HELLP, h*emolytic anemia, *el*evated *l*iver enzymes, and *l*ow *p*latelets.

Adapted from Magee LA, von Dadelszen P, Rey E, et al. Less-tight versus tight control of hypertension in pregnancy. *N Engl J Med*. 2015;372:407–417.

In 2015, CHIPS (Control of Hypertension in Pregnancy Study) added to the evidence base on this topic.[321] CHIPS was an open, international, randomized, controlled clinical trial of tight (target DBP 85 mm Hg) versus less-tight (target DBP 100 mm Hg) BP control in 987 women with mild-to-moderate hypertension in pregnancy. They found no difference in the primary outcome of pregnancy loss or need for high-level neonatal care (Table 48.5). There was no significant difference in the incidence of small-for-gestational-age newborns between the two groups. However, women in the less-tight control group had a higher incidence of severe hypertension, thrombocytopenia, and symptomatic transaminitis. There was a trend toward an increase in HELLP syndrome in the less-tight control group. This study, by far the largest of its kind, provides reassurance that treatment of mild-to-moderate hypertension in pregnancy to a DBP target of 85 mm Hg is safe for the fetus, and lowers risk of short-term maternal pregnancy complications. The study did not report long-term maternal cardiovascular outcomes with tight antihypertensive treatment, but a benefit seems likely given the strong evidence of benefit in nonpregnant persons with chronic hypertension.[322]

GESTATIONAL HYPERTENSION

Gestational hypertension is defined as the new onset of hypertension without proteinuria after 20 weeks' gestation that resolves postpartum. Gestational hypertension likely represents a mix of several underlying etiologies. A subset of women with gestational hypertension have previously existing essential hypertension that is undiagnosed. In such cases, if the woman presents for medical care during the second-trimester nadir in BP, she may be inappropriately presumed to be previously normotensive. In such a circumstance the diagnosis of chronic hypertension is established postpartum, when BP fails to return to normal.

Gestational hypertension progresses to overt preeclampsia in ~10%–25% of cases.[323] When gestational hypertension is severe, it carries similar risks for adverse outcomes as preeclampsia, even in the absence of proteinuria.[324] A renal biopsy study suggests that a significant proportion of women with gestational hypertension have renal glomerular endothelial damage.[142] Hence gestational hypertension may share the same pathophysiologic underpinnings as preeclampsia, and should be monitored and treated as such. In recognition of the syndromic nature of preeclampsia, new American College of Obstetricians and Gynecologists (ACOG) guidelines have eliminated the dependence of proteinuria for the diagnosis of preeclampsia (Table 48.3). In a subset of women with gestational hypertension, it may represent a temporary unmasking of an underlying predisposition toward chronic hypertension. Such women often present with a strong family history of chronic hypertension and develop hypertension in the third trimester with a low uric acid and no proteinuria. Although the hypertension often resolves after delivery, these women are at risk for the development of hypertension and cardiovascular disease later in life.[325,326]

MANAGEMENT OF HYPERTENSION IN PREGNANCY

CHOICE OF AGENTS

Recommendations for the use of antihypertensive agents in pregnancy are summarized in Table 48.6. Methyldopa continues to be the first-line oral agent for the management of hypertension in pregnancy. Methyldopa is a centrally acting alpha-2 adrenergic agonist now seldom used outside of pregnancy. Of all antihypertensive agents, it has the most

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Table 48.6 Safety of Antihypertensive Medications in Pregnancy

Drug	Advantage(s)	Disadvantage(s)
First-Line Agents		
Oral		
Methyldopa	First line; extensive safety data	Short duration of action/multiple daily dosing
Labetalol	Preferred over other β-blockers owing to theoretical beneficial effect of α-blockade on uteroplacental blood flow	Short duration of action/multiple daily dosing. May exacerbate reactive airway disease
Long-acting nifedipine	Once-daily dosing (slow-release preparation)	Edema
Intravenous		
Labetalol	Good safety data	
Nicardipine	Extensive safety data on use as a tocolytic during labor; effective	
Second-Line Agents		
Hydralazine	Extensive clinical experience	Increased risk of maternal hypotension and placental abruption when used acutely
Metoprolol	Potential for once-daily dosing using long-acting formulation	Safety data less extensive than for labetalol
Verapamil, diltiazem	No evidence of adverse fetal effects	Limited data
Generally Avoided		
Diuretics	No evidence of adverse fetal effects	Theoretically may impair pregnancy-associated expansion in plasma volume
Atenolol		May impair fetal growth
Nitroprusside		Risk of fetal cyanide poisoning if used for >4 hours
Contraindicated		
Angiotensin-converting enzyme (ACE) inhibitors		Multiple fetal anomalies; see text
Angiotensin receptor antagonists		Similar risks as ACE inhibitors

extensive safety data and appears to have no adverse fetal effects. Drawbacks include a short half-life, sedation, and rare adverse effects include elevated liver enzymes and hemolytic anemia. Clonidine appears to be comparable with methyldopa in terms of mechanism and safety, but data are fewer.

β-Adrenergic antagonists have been used extensively in pregnancy and are effective without known teratogenicity or known adverse fetal effects. One possible exception is atenolol, which has been associated with fetal growth restriction.[327] Labetalol, which may result in better preservation of uteroplacental blood flow due to its alpha-blocking action, has found widespread use and acceptance, both as an oral and as an intravenous agent.[328]

Calcium channel blockers appear to be safe in pregnancy, and clinical experience with them is growing. Long-acting nifedipine is the most well-studied, and it appears to be safe and effective.[328,329] Nondihydropyridine calcium channel blockers such as verapamil have also been used without apparent adverse effects. However, experience with these agents is more limited than some other classes.

Although diuretics are often avoided in preeclampsia, with the reasoning that circulating volume is already low, there is no evidence that diuretics are associated with adverse fetal or maternal outcomes. Similarly, diuretics are not considered first line in the management of chronic hypertension in pregnancy due to their theoretical impact on the normal plasma volume expansion of pregnancy. When hypertension in pregnancy is complicated by pulmonary edema, diuretics are appropriate and effective.

Angiotensin-converting enzyme inhibitors (ACEIs) and angiotensin receptor antagonists (angiotensin II–receptor blockers or ARBs) are contraindicated in the second and third trimesters of pregnancy. Exposure during this time leads to major fetal malformations including renal dysgenesis, perinatal renal failure, oligohydramnios, pulmonary hypoplasia, hypocalvaria, and IUGR.[330] Evidence for teratogenicity with first-trimester exposure is less compelling. In a large population-based study, Cooper et al[331] reported that congenital malformations of the central nervous and cardiovascular systems were higher among women with first-trimester exposure to ACE inhibitors. However, this study has been criticized for the presence of potential confounders and ascertainment bias. Women with a compelling indication for ACE-I/ARBs (such as diabetic nephropathy) can probably be continued on these agents while attempting conception, with discontinuation as soon as pregnancy is diagnosed. However, risks and benefits of this strategy should be discussed with the patient, with shared and individualized decision making. Women inadvertently exposed in early pregnancy can be reassured by a normal midtrimester ultrasound examination. Fewer data are available on the effects of ARBs,

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but a case series strongly suggests that fetal effects are similar to ACE-I,[332] as would be expected on a theoretical basis.

INTRAVENOUS AGENTS FOR URGENT BLOOD PRESSURE CONTROL

Severe hypertension in pregnancy occasionally requires inpatient management with intravenous agents. There is a lack of controlled studies in humans for the use of all intravenous medications commonly used for urgent control of severe hypertension during pregnancy. Nevertheless, there is extensive clinical experience with several agents, which are widely used with no evidence of adverse effects. Options for intravenous use include labetalol, calcium channel blockers such as nicardipine, and hydralazine.

Intravenous labetalol, like oral labetalol, is safe and effective, with the major drawback being its short duration of action. Intravenous nicardipine has been used safely for tocolysis during premature labor, and recent reports suggest that it is safe in treatment of hypertension as well.[333] The use of short-acting nifedipine is controversial due to well-documented adverse effects in the nonpregnant population. However, a recent meta-analysis suggests that oral short-acting nifedipine can be safely used for severe hypertension during pregnancy,[334] and it may be a good option in areas where intravenous agents are unavailable.

Hydralazine has been widely used as a first-line agent for severe hypertension in pregnancy. However, a meta-analysis of 21 trials comparing intravenous hydralazine with either labetalol or nifedipine for acute management of hypertension in pregnancy suggested an increased risk of maternal hypotension, maternal oliguria, placental abruption, and low Apgar scores with hydralazine.[335] Hence hydralazine should probably be considered second line and its use limited when possible. Nitroprusside carries risk of fetal cyanide poisoning if used for >4 hours and is generally avoided.

ANTIHYPERTENSIVE DRUGS IN BREASTFEEDING

There are few well-designed studies of the safety of antihypertensive medications in breastfeeding women. In general, the agents that are considered safe during pregnancy remain so in breastfeeding. Methyldopa, if effective and well tolerated, should be considered first-line. β-Blockers with high protein binding, such as labetalol and propranolol, are preferred over atenolol and metoprolol, which are concentrated in breast milk.[328] Diuretics may decrease milk production and should be avoided.[328] ACE inhibitors are poorly excreted in breast milk and generally considered safe in lactating women.[336] Enalapril and captopril have been best studied and are preferred in lactating women. Hence in women with proteinuric renal disease, reinitiation of ACE inhibitors should be considered immediately after delivery. Finally, specific data on the pharmacokinetics of each medication should be used to guide mothers to timing breastfeeding intervals before or well after peak breast milk excretion to avoid significant exposure to the baby.

ACUTE KIDNEY INJURY IN PREGNANCY

The incidence of AKI in pregnancy in the developed world has fallen dramatically during the second half of the 20th century.[337] This decline was attributed to improved availability of safe and legal abortion, more widespread and aggressive antibiotic use (both of which decreased the incidence of septic abortion), and improvement in overall prenatal care. However, more recently the incidence of AKI complicating pregnancy has increased in the United States and Canada.[338,339] This trend is likely attributable to the increasing incidence of hypertensive disorders of pregnancy and preeclampsia.[339] Indeed, most cases of AKI in pregnancy in the modern era occur in the setting of hypertensive disorders: gestational hypertension, chronic hypertension, or preeclampsia.[340] Most cases of AKI in pregnancy are mild and self-limited; AKI requiring dialysis is rare, affecting approximately 1:10,000 pregnancies.[341]

Prerenal azotemia—for example, with hyperemesis gravidarum or severe pyelonephritis—is a common cause of AKI in pregnancy. Pregnancy-specific conditions such as preeclampsia, the HELLP syndrome, and AFLP are also common causes of AKI. Obstetric complications such as septic abortion, peripartum sepsis, and placental abruption are associated with severe acute tubular necrosis and rarely bilateral cortical necrosis. Obstructive uropathy is an unusual cause of renal failure in pregnancy.

ACUTE TUBULAR INJURY AND RENAL CORTICAL NECROSIS

Volume depletion complicating hyperemesis gravidarum or uterine hemorrhage (due to placental abruption, placenta previa, failure of the postpartum uterus to contract, or uterine lacerations and perforations) can lead to renal ischemia and subsequent acute tubular injury. AKI may also occur following intraamniotic saline administration, amniotic fluid embolism, and diseases or accidents unrelated to pregnancy.

Renal cortical necrosis is a severe and often irreversible form of acute tubular necrosis that is associated with septic abortion and placental abruption. Septic abortion is an infection of the uterus and the surrounding tissues, most commonly following nonsterile illicit abortions. Septic abortion is now rare where safe therapeutic abortion is available, but remains a serious clinical problem in countries where induced abortion is illegal and/or inaccessible. Women with septic abortion usually present with vaginal bleeding, lower abdominal pain, and fever hours to days after the attempted abortion. Renal failure, which complicates up to 73% of cases,[342] is often characterized by gross hematuria, flank pain, and oligoanuria.

The bacteria associated with septic abortion are usually polymicrobial and derived from the normal flora of the vagina and endocervix, in addition to sexually transmitted pathogens. *Clostridium welchii, Clostridium perfringens, Streptococcus pyogenes,* and gram-negative organisms such as *Escherichia coli* and *Pseudomonas aeruginosa* are all known pathogens. Fatal toxic shock syndrome with *Clostridium sordellii* has been reported following medical termination of pregnancy using mifepristone (RU-486) and intravaginal misoprostol.[343] Pregnancy has long been known to confer a peculiar susceptibility to the vascular effects of gram-negative endotoxin (Shwartzman phenomenon).

AKI with cortical necrosis can also be precipitated by placental abruption. Cortical necrosis can involve the entire renal cortex, often leading to irreversible renal failure, but more commonly is incomplete or patchy. In such cases, a

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protracted period of oligoanuria is followed by a variable return of renal function. The diagnosis of renal cortical necrosis can usually be established by CT scan, which characteristically demonstrates hypodense areas in the renal cortex.

The treatment of AKI in pregnancy is supportive with prompt restoration of fluid volume deficits, and in later pregnancy, expedient delivery. No specific therapy is effective in acute cortical necrosis except for dialysis when needed. Both peritoneal and hemodialysis have been used during pregnancy; however, peritoneal dialysis carries the risk of impairing uteroplacental blood flow.[344] Maternal mortality with pregnancy-associated AKI requiring dialysis remains high, in both developed (4.3%)[341] and developing countries (18.3%).[345]

ACUTE KIDNEY INJURY AND THROMBOTIC MICROANGIOPATHY

The presence of thrombotic microangiopathy (TMA) and acute renal failure in pregnancy is one of the most challenging differential diagnoses to face the nephrologist caring for pregnant patients. Five pregnancy syndromes share many clinical, laboratory, and pathologic features: preeclampsia/HELLP syndrome, TTP/HUS, AFLP, systemic lupus erythematosus (SLE) with the antiphospholipid antibody syndrome, and disseminated intravascular coagulation (usually complicating sepsis). Although it is difficult to establish clinical distinctions between these entities with certainty, the confluence of clinical clues can often establish a likely diagnosis (see Table 48.4).

SEVERE PREECLAMPSIA

AKI affects approximately 1% of women with preeclampsia.[346] However, in patients with severe preeclampsia and HELLP syndrome, the incidence of AKI is much higher, ranging from 10% to 25% in various studies.[347,348] When acute renal failure occurs in the setting of preeclampsia, urgent delivery is indicated.

ACUTE FATTY LIVER OF PREGNANCY

AFLP is a rare but potentially fatal complication of pregnancy, affecting about 1 in 10,000 pregnancies, with a 10% case fatality rate.[349] The clinical picture is dominated by liver failure, with elevated serum aminotransferase levels and hyperbilirubinemia. Severely affected patients have elevations in blood ammonia levels and hypoglycemia. Preeclampsia can also be present in up to half of cases. Hemolysis and thrombocytopenia are not prominent features, and the presence of these findings should suggest the diagnosis of HELLP syndrome or TTP (Table 48.4). Acute renal failure in association with acute fatty liver is seen mainly near term but can occur any time after midgestation.[350] The kidney lesion is mild and nonspecific, and the cause of renal failure is obscure. It may be due to hemodynamic changes akin to those seen in the hepatorenal syndrome or to a TMA.

Although liver biopsy is rarely undertaken clinically, AFLP is a pathologic diagnosis: histologic changes include swollen hepatocytes filled with microvesicular fat and minimal hepatocellular necrosis. This histological picture resembles that seen in Reye syndrome and Jamaican vomiting sickness. A defect in mitochondrial fatty acid oxidation due to mutations in the long-chain 3-hydroxyacyl-coenzyme A dehydrogenase gene has been hypothesized as a risk factor for the development of AFLP.[351] Women with AFLP and their offspring should be tested for this and other genetic defects of fatty acid oxidation, as early diagnosis and treatment of affected neonates may be lifesaving.[352]

Management of AFLP includes supportive care, including aggressive management of the coagulopathy, and prompt termination of the pregnancy. The use of plasmapheresis has been reported both antepartum[353] and postpartum,[354] but there are no controlled trials evaluating its use. The syndrome typically remits postpartum with no residual hepatic or renal impairment, though it can recur in subsequent pregnancies.

THROMBOTIC THROMBOCYTOPENIC PURPURA

TTP is a primary TMA caused by severe ADAMTS13 deficiency. TTP is characterized by thrombocytopenia, hemolysis, and variable organ dysfunction. Acute renal failure may occur, though is not universal.[355] Pregnancy is associated with an increased risk of TTP, usually presenting prior to 24 weeks' gestation,[356] and pregnancy can precipitate relapse in women with a history of TTP.[357] Deficiency in the von Willebrand factor cleaving protease (ADAMTS13) has been linked to the pathogenesis of TTP in nonpregnant states, but this has not been well-studied in pregnancy. ADAMTS13 levels fall during the second and third trimester, potentially contributing to the increased incidence of TTP in pregnancy.[358]

The often challenging clinical distinction between TTP and preeclampsia/HELLP is important for patient management, because plasma exchange is beneficial in TTP but not in the HELLP syndrome. A history of preceding proteinuria, hypertension, and severe liver injury is more suggestive of the HELLP syndrome, whereas the presence of renal failure and severe nonimmune hemolytic anemia is more typical of TTP (Table 48.4). Although plasmapheresis for HUS/TTP in pregnancy and postpartum has not been evaluated in controlled studies, case series suggest that it is safe and effective.[357] Termination of pregnancy does not appear to alter the course of HUS/TTP, thus it is not generally recommended unless the fetus is compromised.

HEMOLYTIC UREMIC SYNDROME

Atypical HUS, can cause the abrupt onset of acute renal failure, often in the peripartum or early postpartum period.[359] This disorder is due to a hereditary or acquired deficiency of complement regulatory proteins, leading to activation of the alternative complement pathway. Most of the reports of "HUS" in pregnancy predated the discovery of the role of complement pathway dysregulation in the pathogenesis of this disorder. In one series of 21 patients with pregnancy-associated HUS, 57% had low C3, and 86% of women had a detectable complement gene mutation; the most common abnormalities were CFH mutation (45%), CFI mutation (9%), and C3 mutation (9%).[360] No patients in this series had antibody-mediated disease. Long-term renal outcomes in the pre-eculizumab era were poor, with the majority of women progressing to ESRD.[360] Although data are limited, there are reports of the successful use of eculizumab for the treatment of complement-mediated TMA in pregnancy, without overt adverse fetal effects.[361] Unlike preeclampsia, delivery does not alter the course of TTP or complement-mediated TMA.

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OBSTRUCTIVE UROPATHY AND NEPHROLITHIASIS

AKI due to bilateral ureteral obstruction is a rare complication of pregnancy. Because of the physiologic hydronephrosis of pregnancy (see the "Physiologic Changes of Pregnancy" section), the diagnosis of urinary obstruction can be challenging. If clinical suspicion is high (e.g., marked hydronephrosis, abdominal pain, elevated serum creatinine), a percutaneous nephrostomy may be needed as a diagnostic and therapeutic trial to confirm the diagnosis of obstructive uropathy. If present, obstruction can be managed with ureteral stenting[362] or delivery.[363]

Circulating levels of 1,25-dihydroxyvitamin D_3 are increased during normal pregnancy, resulting in increased intestinal calcium absorption. Urinary excretion of calcium is also increased, leading to a tendency in some women to form kidney stones. Excessive intake of calcium supplements can lead to hypercalcemia and hypercalciuria. Although intestinal absorption and urinary excretion of calcium are increased, there is no evidence that the risk of nephrolithiasis is increased, possibly due to a concomitant increase in urine flow and physiological dilation of the urinary tract.

Calcium oxalate and calcium phosphate constitute the majority of the stones produced during pregnancy. As in nonpregnant patients, ureteral calculi in pregnancy produce flank pain and lower abdominal pain with hematuria. Premature labor is sometimes induced by the intense pain, and the risk of infection is increased. Ultrasonography and MRI are the preferred methods to exclude obstruction and visualize stones.[364] The management of renal calculi is conservative with adequate hydration, analgesics, and antiemetics. Thiazide diuretics and allopurinol are avoided during pregnancy. Twenty-four-hour urine collection to quantify urinary calcium and uric acid excretion is recommended after delivery. Nephrolithiasis complicated by urinary tract infection should be treated with antibiotics for 3–5 weeks, followed by suppressive treatment after delivery, because the calculus may serve as a nidus of infection. Most stones pass spontaneously, but placement of a ureteral stent may be required if ureteral obstruction occurs.[365] Lithotripsy is relatively contraindicated during pregnancy due to its adverse effects on the fetus. However, extracorporeal shock wave lithotripsy has been used during the first 4–8 weeks of pregnancy without known adverse consequences to the fetus.[366]

URINARY TRACT INFECTION AND ACUTE PYELONEPHRITIS

Infections of the urinary tract represent the most frequent renal problem encountered during gestation.[367] Although the prevalence of asymptomatic bacteriuria—which ranges between 2% and 10%—is similar to that in nonpregnant populations, it needs to be managed more aggressively for several reasons. Urinary stasis predisposes pregnant women to ascending pyelonephritis in the setting of cystitis. Hence while asymptomatic bacteriuria in the nonpregnant state is usually benign, untreated asymptomatic bacteriuria in pregnancy can progress to overt cystitis or acute pyelonephritis in up to 40% of patients.[368] Acute pyelonephritis is a serious complication during pregnancy, usually presenting between 20 and 28 weeks of gestation with fevers, loin pain, and dysuria. Sepsis resulting from pyelonephritis can progress to endotoxic shock, disseminated intravascular coagulation, acute renal failure, and respiratory failure.[368] Asymptomatic bacteriuria has also been associated with an increased risk of premature delivery and low birth weight.[369] Treatment of asymptomatic bacteriuria during pregnancy has been shown to reduce these complications and improve perinatal morbidity and mortality.[370] Thus early detection and treatment of asymptomatic bacteriuria are warranted.

The usual signs and symptoms of urinary tract infection can be unreliable in pregnancy. Dysuria and urinary frequency are common during the latter half of pregnancy in the absence of infection, due to pressure on the bladder from the gravid uterus. Low-grade pyuria is often present because of contamination by vaginal secretions. The use of the urinary dipstick to screen for bacteriuria is associated with a high false-negative rate and quantitative urine culture is preferred for screening. More than 10^5 bacteria/mL of a single species indicates significant bacteriuria. Screening for asymptomatic bacteriuria is recommended during the first prenatal visit and is only repeated in high-risk women such as those with a history of recurrent urinary infections or urinary tract anomalies.

If asymptomatic bacteriuria is found, prompt treatment is warranted (usually with a cephalosporin) for at least 3–7 days.[371] Treatment with a single dose of fosfomycin has also been used successfully. Trimethoprim-sulfa and tetracycline are contraindicated in early pregnancy because of their association with birth defects. A follow-up culture 2 weeks after treatment is necessary to ensure eradication of bacteriuria. Suppressive therapy with nitrofurantoin or cephalexin is recommended for those patients with bacteriuria that persists after two courses of therapy.[372] Prolonged suppressive treatment of bacteriuria has been shown to reduce the incidence of pyelonephritis.[373] Because of the high maternal morbidity and mortality associated with pyelonephritis, it is usually treated aggressively with hospitalization, intravenous antibiotics, and hydration. Infections of the urinary tract are further discussed in Chapter 36.

CHRONIC KIDNEY DISEASE IN PREGNANCY

Women who enter pregnancy with CKD are at increased risk for adverse maternal and fetal outcomes, including rapid decline of renal function and perinatal mortality. Although the frequency of live births now exceeds 90% in these women, the risks for preterm delivery, IUGR, perinatal mortality, and preeclampsia are significantly elevated.[374,375] Preterm delivery results in both immediate and long-term morbidity for the offspring, with an increased risk of cardiovascular and renal disease later in life.[376] The goals for pregnancy in women with CKD are to preserve maternal renal function during and after pregnancy, and to maximize the likelihood of successful term or near-term delivery for the fetus.

The physiologic increase in renal blood flow and GFR characteristic of normal pregnancy is attenuated in chronic renal insufficiency.[375] The stress of greater renal blood flow during pregnancy may exacerbate renal damage in the setting of preexisting renal disease, similar to nonpregnant states in which the impaired kidney is more sensitive to such insults.

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Indeed, worsening of hypertension and proteinuria are common during pregnancy if these conditions exist prior to pregnancy,[377] and in concert with these observations, overall maternal and fetal prognosis correlates with the degree of hypertension, proteinuria, and renal insufficiency prior to conception.

Fortunately, there is good evidence to suggest that women with underlying kidney disease but only mild renal impairment, normal BP, and no proteinuria have good maternal and fetal outcomes, with little risk for accelerated progression toward ESRD.[375,378] However, even women with stage 1 CKD are at increased risk of cesarean section, preterm delivery, and the need for neonatal intensive care as compared with low-risk control pregnancies.[379] Although debate exists whether specific renal diseases are more commonly associated with an accelerated decline in renal function, data suggest that the degree of renal insufficiency, presence and severity of hypertension, and severity of proteinuria, rather than the underlying renal diagnosis, are the primary determinants of outcome.[379] Women who become pregnant with a serum creatinine >1.4–1.5 mg/dL are more likely to experience a decline in renal function than women with a comparable degree of renal dysfunction who do not become pregnant.[375] Initiating pregnancy with a serum creatinine >2.0 mg/dL carries a high risk (>30%) for accelerated decline in renal function both during and after pregnancy.[374] Furthermore, among women with a serum creatinine >2.5 mg/dL, over 70% experience preterm delivery, and over 40% experience preeclampsia.[374,377] Measures to predict which women will experience rapid decline postpartum do not exist, and terminating pregnancy does not reliably reverse the decline in renal function. Recent data in women with autosomal dominant polycystic kidney disease and normal renal function suggest that there is a high rate of successful uncomplicated pregnancies.[380] Other specific conditions including diabetic nephropathy and lupus nephritis are discussed later. Regardless of cause of renal disease, the tenet that an elevated serum creatinine >1.4–1.5 mg/dL puts women at increased risk for renal decline holds true. An overall approach to preconception counseling and prenatal care of women with CKD is shown in Fig. 48.14.

DIABETIC NEPHROPATHY

The incidence and prevalence of diabetes throughout the world is increasing,[381] and the number of women entering pregnancy with diabetes is rising.[382] However, the incidence of diabetic nephropathy among pregnant women with diabetes has been declining in recent years,[383,384] likely due to improvements in glycemic control, hypertension management, and more widespread use of inhibitors of the renin–angiotensin–aldosterone system among young women with diabetes.

Women with pregestational diabetes, with or without nephropathy, have a higher risk of adverse fetal and maternal outcomes, including congenital malformation, preterm birth, macrosomia, preeclampsia, and perinatal mortality.[44,385] The presence of albuminuria confers a particularly high risk for preeclampsia and preterm delivery.[386] Poor glycemic control before and during pregnancy is strongly linked to adverse pregnancy outcomes, including preeclampsia and serious adverse fetal outcomes; hence, endocrine consultation before pregnancy is strongly advised.[387,388]

Pregnancy itself does not appear to accelerate the progression of kidney disease if kidney function is normal or near normal prior to pregnancy.[389] The prognosis changes, however, if renal function is impaired at pregnancy onset. Compared with preconception measures, creatinine clearance is notably lower even within the first few months postpartum in women initiating pregnancy with impaired renal function.[390] In a study of 11 patients with diabetic nephropathy and serum creatinine >1.4 mg/dL at pregnancy onset, >40% progressed to end-stage renal failure within 5–6 years after pregnancy.[391] Aggressive BP control before and after pregnancy may attenuate the postpartum decline in renal function.[389] Nevertheless, inexorable decline in renal function following pregnancy is common in women initiating pregnancy with diabetic nephropathy and impaired renal function. For this reason, women with diabetic nephropathy are usually advised to postpone pregnancy until after renal transplantation, which improves fertility status and fetal outcomes.

ACEIs and ARBs are contraindicated in the second and third trimesters of pregnancy, leading to birth defects including hypocalvaria (hypoplasia of the membrane bones of the skull), renal tubular dysplasia, and IUGR.[328] The inability to use these medications may contribute to progression of renal failure in pregnant women with diabetic nephropathy and impaired renal function. However, population-based studies find no adverse fetal effects with first-trimester ACEI and ARB exposure.[328,392,393] Unfortunately, it is difficult to exclude first-trimester teratogenicity completely: for example, a case of exencephaly and unilateral renal agenesis was reported in a fetus of a mother who was taking an ARB at conception.[394] Some experts suggest an approach whereby women with a strong indication for ACEI or ARB, such as those with diabetic nephropathy, be continued on these medications while attempting conception, with instructions to discontinue them as soon as pregnancy is recognized. This approach minimizes both fetal risk and the amount of time that women must be off these agents which have clear maternal benefit.

The diabetic milieu during pregnancy may also subsequently affect metabolism and renal function in the offspring.[395,396] For example, in a cross-sectional study of 503 Pima Indians with type 2 diabetes, the prevalence of albuminuria was significantly higher in the offspring of mothers with diabetes during pregnancy (58%) compared with offspring of mothers without evidence of diabetes during pregnancy (40%).[396] It is speculated that abnormal in utero exposure in the offspring of diabetic mothers leads to impaired nephrogenesis and reduced nephron mass, and this puts the offspring at higher risk for developing renal disease and hypertension in later life (see also Chapter 21).

LUPUS NEPHRITIS AND PREGNANCY

SLE is common in women of childbearing age. During pregnancy, women with SLE are at increased risk for preterm birth, IUGR, spontaneous abortions, and preeclampsia.[397,398] The presence of superimposed renal disease increases the risk of these complications even further.[399,400] Over the past few decades, improvements in disease management and perinatal monitoring have led to a decrease in pregnancy loss and preterm deliveries.[401] Outcomes, however, still appear to be poor in developing countries.[402] With careful planning, monitoring, and management, the majority of patients with

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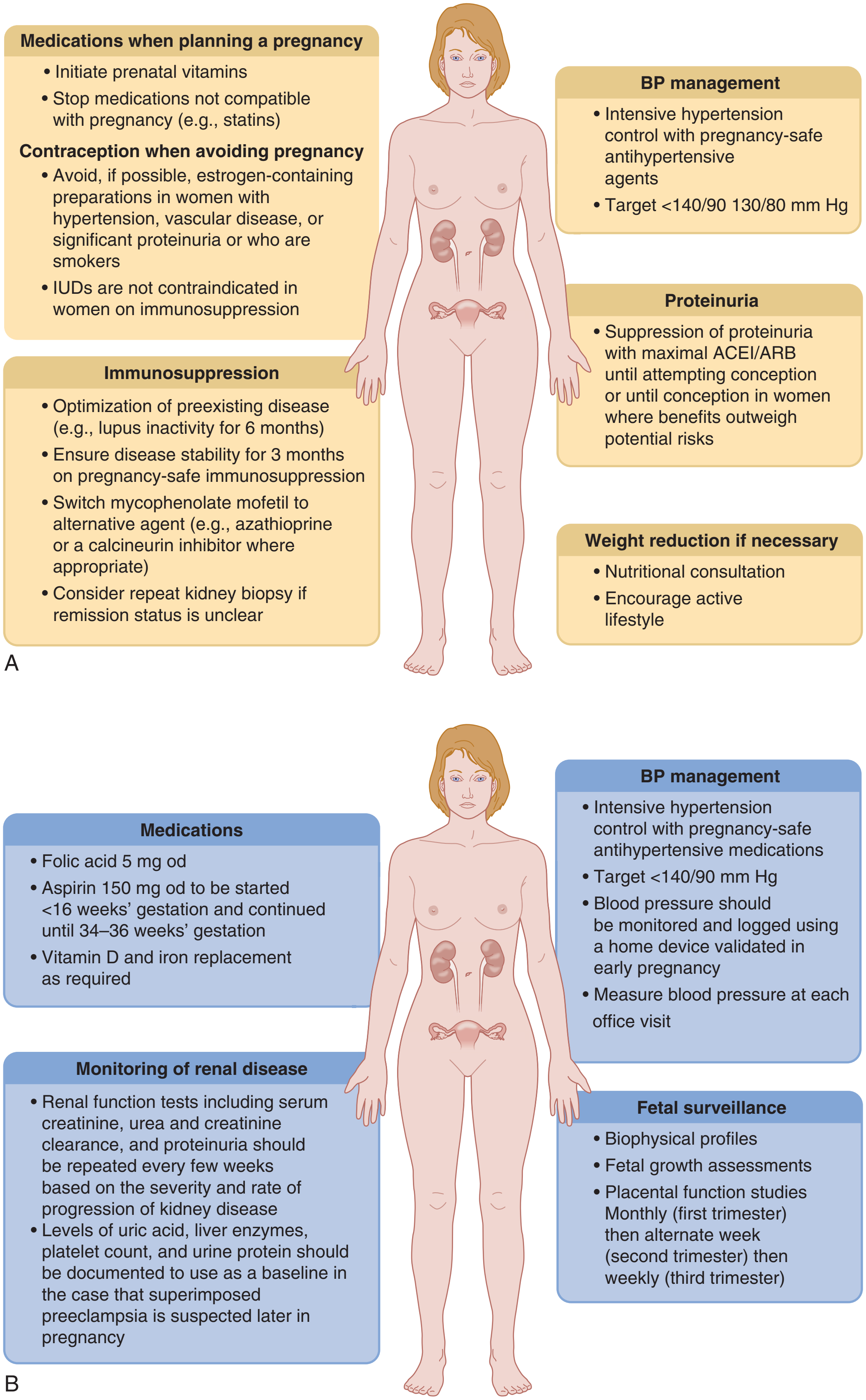

Fig. 48.14 (A) Prepregnancy optimization in women with chronic kidney disease. (B) Antenatal care in women with chronic kidney disease. *ACEI,* Angiotensin-converting enzyme inhibitor; *ARB,* angiotensin receptor blocker; *BP,* blood pressure; *IUD,* intrauterine device; *od,* once daily. (Figure adapted from Hladunewich MA, Melamad N, Bramham K. Pregnancy across the spectrum of chronic kidney disease. *Kidney Int.* 2016;89:995–1007.)

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SLE—especially those with normal baseline renal function—can complete pregnancy without serious maternal or fetal complications.[403]

Among pregnant women with SLE, risk factors for adverse pregnancy outcomes include lupus disease activity, history of lupus nephritis, antihypertensive use, the presence of antiphospholipid antibodies, thrombocytopenia, and nonwhite or Hispanic race.[398,404,405] Proliferative [World Health Organization (WHO) class III or IV] lupus nephritis is associated with a higher risk of preeclampsia and lower birth weight than mesangial (WHO class II) or membranous (WHO class V) lupus nephritis.[406] Alterations in circulating angiogenic factors (high sFlt1 and low PlGF) measured in the first half of pregnancy are strongly associated with adverse pregnancy outcomes, including preterm preeclampsia, fetal/neonatal death, and indicated preterm delivery < 30 weeks.[407] In the future, these may prove to be useful biomarkers for risk stratification in women with lupus in pregnancy.

Active nephritis at conception (defined in most studies as proteinuria > 500 mg/day and/or an active urinary sediment) is also a risk factor for preterm birth.[404,405] Hence, women should postpone pregnancy until lupus activity is quiescent for ~6 months and immunosuppressives are minimized.[408,409] Mycophenolate mofetil (MMF) is teratogenic and should be switched to azathioprine at least 3 months prior to conception; this approach is associated with a low risk of renal flares.[410]

Approximately 10%–15% of women with a history of lupus nephritis experience a nephritis flare during pregnancy.[411,412] De novo lupus nephritis in pregnancy is less common, affecting only 2% of women with SLE but without renal disease prior to pregnancy. Low C4, but not low C3 or anti-double-stranded DNA antibody positivity, is predictive of pregnancy-associated lupus nephritis flare.[411] Prophylactic therapy with steroids does not appear to prevent a lupus flare during pregnancy.[413] Discontinuation of hydroxychloroquine in pregnancy is associated with an increased risk of lupus flare, so this agent should generally be continued.[414] Steroids, azathioprine, and hydroxychloroquine may be used to manage lupus nephritis flares, and treat extrarenal lupus symptoms, during pregnancy.[415]

Pregnant women with SLE are at risk for both lupus nephritis flare and preeclampsia. Unfortunately, both syndromes share the common presenting symptoms of hypertension and proteinuria, hence distinguishing the two can be a clinical challenge. The distinction between preeclampsia and lupus flare is critical, however, especially in women presenting prior to 37 weeks' gestation because the treatments differ. For a lupus nephritis flare, steroids, azathioprine, and CNIs may quell the disease, allowing pregnancy to continue, whereas for preeclampsia, induction of delivery is the definitive treatment. Even in the medical literature, reports of preeclampsia in the setting of lupus renal disease are prone to misclassification, because the diagnosis of preeclampsia is usually based on hypertension and proteinuria, and not on renal biopsy or specific serologic testing. Alterations in components of the complement cascade (e.g., reductions in C3, C4, CH50) may be useful to help identify pregnant women with a lupus flare, and to distinguish lupus flare from preeclampsia.[416] An active urinary sediment suggests lupus nephritis, as the sediment in preeclampsia is typically bland. However, none of these clinical findings are definitive and kidney biopsy during pregnancy is sometimes indicated (see the "Kidney Biopsy in Pregnancy" section). On the horizon, measurement of circulating angiogenic factors may aid clinical decision making, especially during the critical period before term.[76,417,418]

Treatment of severe proliferative lupus nephritis in pregnancy is challenging, as conventional induction therapies are contraindicated due to risk for congenital malformations (mycophenolate[419]) and fetal loss (cyclophosphamide[420]). Prednisone is safe, though the risk of gestational diabetes is increased.[408] CNIs are nonteratogenic and can be used to treat lupus nephritis in pregnancy, with some data to support their efficacy in the nonpregnant population.[408] Rituximab lacks safety data in pregnancy, and it is not recommended.[421] Azathioprine is considered relatively safe in pregnancy, and is commonly used as adjunctive or maintenance therapy in pregnant patients with lupus nephritis.[415]

KIDNEY BIOPSY IN PREGNANCY

Indications for kidney biopsy during pregnancy are similar to those in the nonpregnant population; specifically, clinical suspicion of glomerular disease where pathologic diagnosis is likely to change management. The best data on the safety of kidney biopsy during pregnancy are from a 2013 systematic review of 243 kidney biopsies performed during pregnancy, and 1236 kidney biopsies performed during the postpartum period.[422] Kidney biopsies performed during pregnancy had a higher complication rate than biopsies performed after pregnancy (7% vs. 1%). However, most complications were minor: Only four cases of major bleeding occurred, all in women who were biopsied between 23 and 26 weeks of gestation. For this reason, and because this is a period of uncertain fetal viability should complications require preterm delivery, kidney biopsy should be avoided between 23 and 26 weeks' gestation. Later in gestation, kidney biopsy becomes technically difficult, and often can be postponed until after delivery.

PREGNANCY IN END-STAGE KIDNEY DISEASE

End-stage kidney disease is characterized by severe hypothalamic–pituitary–gonadal dysfunction, which is reversed by transplantation but not by conventional dialysis. Women of childbearing age on dialysis frequently have menstrual disturbances, anovulation, and infertility.[423] Animal studies suggest that uremia impairs fertility via aberrant neuroendocrine regulation of hypothalamic gonadotropin-releasing hormone secretion.[424] Gonadal function is impaired in men as well, who can experience testicular atrophy, hypospermatogenesis, infertility, and impotence. However, in recent decades the Toronto experience has suggested that fertility in women with ESRD may be improved by intensive hemodialysis (≥36 hours/week or more).[425]

Conception on dialysis is unusual, but not impossible. Hence contraception remains important in women of childbearing age who do not wish to become pregnant. In fact, pregnancy and live birth rates have been increasing in young women receiving dialysis.[426] Survey data from Japan, Belgium, and the United States from the 1990s reported pregnancy rates among women of childbearing age with ESRD of 0.3% to 2%.[375,427–430] In these studies, approximately half of pregnancies resulted in successful delivery of a live infant, though the majority (84% in the 1994 study) were premature.

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More recent reports of pregnancies in the setting of chronic hemodialysis have reported somewhat higher neonatal survival (70%–75%).[431–433] Live birth rates of up to 86% have been reported with intensive dialysis (see later).[434] Adverse fetal outcomes are most often due to preterm labor, premature rupture of membranes, polyhydramnios, and IUGR. The likelihood of successful pregnancy is higher in women who conceive prior to initiation of dialysis, as compared with women who conceive after initiation of dialysis.[433]

When pregnancy does occur in a woman on chronic hemodialysis, a large increase in dialysis time appears to improve pregnancy outcomes.[435] Initial reports suggested that women who received ≥20 hours/week of hemodialysis had improved neonatal outcomes and longer gestations.[428,429,436] In 2014, Hladunewich and colleagues[434] pushed the envelope even further, reporting an impressively high live birth rate of 86.4% and mean gestational age at delivery of 36 weeks among women who received 43 ± 6 dialysis hours/week, compared with a live birth rate of 61.4% and mean gestational age at delivery of 27 weeks among historical control women who received 17 ± 5 dialysis hours/week. They observed a dose–response relationship between dialysis dose and pregnancy outcomes, with the highest live birth rate and longest gestation among women who received ≥37 hours/week of hemodialysis. This extremely high dialysis intensity requires a major commitment on the part of the patient and the dialysis provider, and is most realistically attained by daily nocturnal dialysis.

Volume management is challenging, as the dry weight increases up to 0.5 kg/week throughout pregnancy, and hypovolemia needs to be vigilantly avoided. Medications must be carefully reviewed so as to avoid drugs toxic to the fetus, such as ACE inhibitors. Erythropoietin and iron requirements increase in pregnancy, and dosing should be adjusted to approximate the physiologic anemia of pregnancy (10–11 g/L), as low hematocrit has been associated with adverse fetal outcomes.[437] Exacerbation of hypertension is common, though the incidence of preeclampsia is difficult to ascertain due to the inability to apply standard diagnostic criteria. If preeclampsia occurs, intravenous magnesium for seizure prophylaxis should be dosed cautiously, as magnesium excretion is impaired in ESRD. Magnesium levels should be measured frequently, and patients should be monitored for signs of toxicity such as hyporeflexia, weakness, and sedation. Close monitoring of fetal growth and well-being, in collaboration with an obstetrician, is essential after 24 weeks' gestation as IUGR and fetal distress are common.

Data on pregnancy outcomes in women on peritoneal dialysis are limited to case reports and small case series. Reports suggest a relatively high rate of complications such as peritonitis, catheter malfunction, and exit site infection.[429,438,439] Peritoneal dialysis may be associated with a higher rate of small-for-gestational-age infants, as compared with hemodialysis.[435] Supplementation of peritoneal dialysis with intermittent hemodialysis has been reported as a strategy to increase dialysis dose in pregnancy.[440]

PREGNANCY IN THE KIDNEY TRANSPLANT RECIPIENT

While women with ESRD on dialysis are often infertile, kidney transplantation results in a return to normal hormonal function and fertility within 6 months in approximately 90% of women of childbearing age.[441] Because pregnancy in advanced CKD is associated with a high risk of pregnancy complications and accelerated loss of kidney function, women with advanced CKD are often counseled to delay pregnancy until after kidney transplantation, when pregnancy outcomes are more favorable. Over 14,000 pregnancies in renal allograft recipients have been documented since 1958 (Fig. 48.15).[442] However, the pregnancy rate among female kidney transplant recipients aged 20–45 has declined since 1990 in both the United States and Australia/New Zealand.[443,444] Reasons for this decline are speculative, but may include avoidance of pregnancy in patients receiving mycophenolate immunosuppression (introduced in 1995), and more advanced maternal age and comorbidities among more recent transplant recipients.

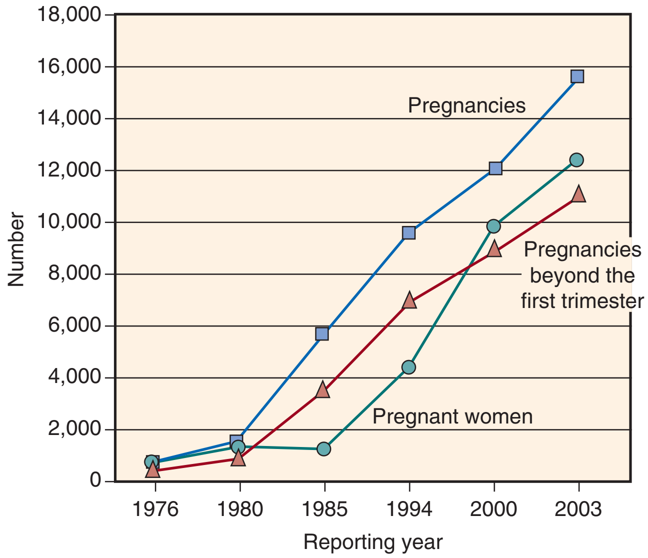

Fig. 48.15 Pregnancies in kidney transplant recipients worldwide. The *circles* represent the numbers of pregnancies reported worldwide in kidney transplant recipients during the indicated year. The *numbers* include therapeutic terminations, spontaneous abortions, ectopic pregnancies, and stillbirths. The *squares* represent the numbers of transplant recipients reported to have been pregnant during that year, again including all outcomes. The *triangles* represent the numbers of pregnancies beyond the first trimester reported in the literature during the indicated year. The data are from the US National Transplantation Pregnancy Registry, the European Dialysis and Transplant Association Registry, and the UK Transplant Pregnancy Registry. (From McKay DB, Josephson MA. Pregnancy in recipients of solid organs—effects on mother and child. *N Engl J Med.* 2006;354:1281–1293.)

Although the majority of pregnancies following kidney transplantation lead to excellent outcomes for both mother and fetus, such pregnancies are not without risk and require close monitoring and collaboration between the nephrologist and the obstetrician. The goals of care in these patients are to optimize maternal health, including graft function and hypertensive disorders of pregnancy, and to maximize the likelihood of a healthy newborn.

FETAL AND NEONATAL OUTCOMES

Four major registries, the US National Pregnancy Transplantation Registry,[445] the European Dialysis and Transplant Association Registry,[446] the UK Transplant Pregnancy

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Registry,[447] and the Australia and New Zealand Dialysis and Transplant Registry,[448] have documented pregnancy outcomes on >2600 pregnancies in women with solid organ transplants. In addition, a 2011 meta-analysis summarized 50 studies on pregnancy outcomes from >4700 pregnancies in transplant recipients.[449] Statistics on the major complications of pregnancy from these studies are remarkably consistent. Approximately 20% of pregnancies among renal transplant recipients end in the first trimester, half due to miscarriage and the remainder due to elective termination. The live birth rate is 73%–79%; however, there is a substantial risk of low birth weight (40%–54%) and preterm delivery (45%–52%). Ectopic pregnancy appears to be slightly increased, especially in pregnancies that occur soon after transplant, but the rate remains below 1%.[449] The rate of structural birth defects is no higher than the general population. Vaginal delivery is safe, and cesarean section should be performed only for obstetrical indications.

TIMING OF PREGNANCY AFTER TRANSPLANTATION

Pregnancy within the first 6–12 months following transplantation is undesirable for several reasons. The risk of acute rejection is relatively high, immunosuppressant medications are at higher dosages, and risk of infection is greatest.[442] Traditionally it has been recommended that women wait ~2 years after transplant prior to attempting conception.[450] However, many kidney transplant recipients are of advanced maternal age, hence delaying pregnancy may lead to age-related decrease in fertility. The American Society of Transplantation suggests that conception could be safely considered as early as 1 year posttransplant for women on stable doses of nonteratogenic immunosuppressive agents, with normal renal function (Cr < 1.5 mg/dL and urine protein excretion of <500 mg/day), no concurrent fetotoxic infections (such as cytomegalovirus), and with no rejection episodes within the past year.[442]

EFFECT OF PREGNANCY ON RENAL ALLOGRAFT FUNCTION

Pregnancy itself does not appear to adversely affect graft function in transplant recipients, provided baseline graft function is normal and significant hypertension is not present.[450] In general, when pregnancy occurs 1–2 years after transplant, the rejection rate is similar to that seen in non-pregnant controls (3%–4%).[449,451] When moderate renal insufficiency is present (serum creatinine >1.5–1.7 mg/dL), pregnancy does carry a risk of progressive renal dysfunction,[452] as well as an increased risk of a small-for-gestational-age infant and of preeclampsia.[453]

Only two small case–control studies have reported long-term (>10 years) graft function after pregnancy. One study suggested that 10-year graft survival may be diminished in transplant recipients who become pregnant compared with controls who did not become pregnant.[454] A second study reported no significant difference.[455] Further studies reporting long-term outcomes in the era of CNIs are needed.

Due to ongoing immunosuppression, transplant recipients are at risk for infections that have implications for the fetus, including cytomegalovirus, herpes simplex, and toxoplasmosis. The rate of bacterial urinary tract infections is increased (~13%–40%),[452] but these are usually treatable and uncomplicated.

The most common complication of pregnancy in transplant recipients is hypertension, which affects between 30% and 75% of pregnancies among transplant recipients.[449,451,456] Hypertension is likely due to a combination of underlying medical conditions and the use of CNIs. Preeclampsia complicates 25%–30% of pregnancies in kidney transplant patients,[449,451,452,457] and diagnosis is often challenging due to the frequent presence of hypertension and/or proteinuria at baseline. The American Society of Transplantation recommends that hypertension in pregnant renal transplant recipients should be managed aggressively, with target BP close to normal – a goal which differs from somewhat higher BP goals in women with hypertension in pregnancy in the absence of a transplant.[328,442] Agents of choice (Table 48.6) include methyldopa, nonselective β-adrenergic antagonists (e.g., labetalol), and calcium channel blockers. ACE inhibitors and angiotensin receptor blockers should be avoided in pregnancy, particularly in the second and third trimesters. Details about the use of these agents in pregnancy are discussed in greater detail in the section Chronic Hypertension and Gestational Hypertension.

MANAGEMENT OF IMMUNOSUPPRESSIVE THERAPY IN PREGNANCY

The US Food and Drug Administration replaced the former pregnancy risk letter categories on prescription and biological drug labeling with new information to make them more meaningful to both patients and health-care providers.[458] The benefit-to-risk ratio for most immunosuppressive agents used in the treatment of transplant recipients favors medication continuation during pregnancy. Certain immunomodulators, however, can cause extreme fetal harm and should be used with caution. There is a significant amount of published data that can inform decisions to use these agents safely in pregnancy (Table 48.7).

CNIs (either cyclosporine or tacrolimus) and steroids, with or without azathioprine, form the basis of immunosuppression during pregnancy. Corticosteroids at low to moderate doses (5–10 mg/day) are safe and prednisone is therefore the agent of choice during pregnancy.[459] Stress-dose steroids are indicated in the peripartum period. Azathioprine is generally considered safe at dosages <2 mg/kg per day. Higher doses of azathioprine are associated with congenital anomalies, immunosuppression, and IUGR, and should be avoided if possible.[450]

CNIs are widely used in pregnancy, and generally considered safe.[459–461] Clinical data have not demonstrated an increased incidence of congenital malformations, with the possible exception of low birth weight.[459] Owing to decreased gastrointestinal (GI) absorption, increased volume of distribution, and increased renal clearance, dosages of CNIs generally need to be increased in pregnancy to maintain therapeutic levels, and close monitoring of drug levels is essential.[462,463]

Sirolimus causes fetal toxicity in animal studies, and is generally contraindicated in pregnancy.[459] Whenever possible, sirolimus should be discontinued at least 12 weeks prior to conception. The risk of fetal malformations is highest at 30–71 days' gestation, so there is a window to stop sirolimus if pregnancy is detected early. Nevertheless, sirolimus should be discontinued preemptively in women of childbearing age who are not using contraception.

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Table 48.7 Immunosuppressive Medications in Pregnancy

Drug	Recommendations
Prednisone	Safe when used long term at low to moderate doses (5–10 mg/day) Safe when given acutely at high doses for treatment of acute rejection
Cyclosporine	Extensive clinical data suggest safe at low-to-moderate clinical doses Changes in absorption and metabolism require close monitoring of levels and frequent dose adjustments in pregnancy
Tacrolimus	Similar to those for cyclosporine, although somewhat less data available
Sirolimus	Embryo/fetal toxicity in rodents was manifested as mortality and reduced fetal weights (with associated delays in skeletal ossification); human studies are lacking
Azathioprine	Considered safe at dosages < 2 mg/kg per day, but at higher doses associated with fetal growth restriction
Mycophenolate mofetil	Contraindicated in pregnancy (teratogenic in animal and human studies)
Muromonab-CD3 (OKT-3)	Case reports of successful use for induction in unsuspected pregnancy and for acute rejection, but data are limited
Antithymocyte globulin (C)	Animal studies have not been reported, and there are no controlled data from use in human pregnancy
Belatacept, basiliximab, alemtuzumab	Avoid in pregnancy; inadequate safety data in humans

Adapted from McKay DB, Josephson MA. Pregnancy after kidney transplantation. *Clin J Am Soc Nephrol.* 2008;3:S117–S125.

MMF is associated with developmental toxicity, malformations, and intrauterine death in animal studies at therapeutic dosages. Human data are limited to isolated case reports, but suggest that MMF may be associated with spontaneous abortions and with major fetal malformations,[464] especially in combination with cyclosporine.[459] Hence MMF, like sirolimus, should be avoided in pregnancy. For a more detailed discussion of data on effects of these agents on neonatal immunologic function and long-term outcomes, the reader is referred to reviews by Josephson and McKay.[465,466]

MANAGEMENT OF ACUTE REJECTION IN PREGNANCY

The incidence of acute rejection during pregnancy is approximately 4%, similar to the nonpregnant population.[449] However, the diagnosis of acute rejection during pregnancy can be difficult. Acute rejection should be suspected if fever, oliguria, graft tenderness, or deterioration in renal function are noted. Biopsy of the renal graft should be performed to confirm the diagnosis prior to initiation of treatment. High-dose steroids are the mainstay of treatment of acute rejection during pregnancy.[450,467] Little data are available on the safety of agents such as OKT-3, antithymocyte globulin, belatacept, alemtuzumab, daclizumab, and basiliximab in pregnancy, and they are generally avoided (Table 48.7).[459] The National Transplantation Pregnancy Registry reported five cases of OKT-3 use during pregnancy, with four surviving infants.[451] Polyclonal and monoclonal antibodies would be expected to cross the placenta, but fetal effects are largely unknown.

BREASTFEEDING AND IMMUNOSUPPRESSIVE AGENTS

Studies on transfer of CNIs to the babies of breastfeeding mothers are inconsistent, with some studies reporting undetectable levels,[459,468] and others reporting high neonatal blood concentrations.[469] There are no reports on adverse neonatal effects with cyclosporine or tacrolimus use in breastfeeding. Limited data suggest that tacrolimus levels are very low in breast milk. Investigators from the National Transplantation Pregnancy Registry suggest that breastfeeding should not be discouraged in women taking tacrolimus.[470] When cyclosporine or tacrolimus is used in a lactating mother, consideration should be given to monitoring neonatal blood concentrations for potential toxicity. Theoretically, MMF should be safe in breastfeeding because the active metabolite secreted in breast milk, MMA, is not GI bioavailable; however, human evidence of safety is lacking. Nevertheless, breastfeeding among transplant recipients remains controversial.[466]

PREGNANCY OUTCOME FOLLOWING KIDNEY DONATION

Gestational hypertension and preeclampsia are more common among living kidney donors as compared with matched nondonors.[471] There are conflicting data on whether adverse pregnancy outcomes, such as preterm delivery and fetal loss, are also increased among kidney donors.[472,473] Women of reproductive age who are considering kidney donation should be counseled regarding these potential risks.

CONCLUSION

Recent decades have brought significant advances in our understanding of the pathogenesis of preeclampsia. We have more data than ever before to guide management of hypertension, CKD, and glomerular disease in pregnancy. Nevertheless, large evidence gaps remain with regard to BP targets and use of novel immunosuppressive and biologic agents in women with hypertension and kidney disease in pregnancy. Preeclampsia continues to cause major maternal and neonatal morbidity and mortality worldwide, as delivery remains the only effective treatment. In the future, we look toward better and larger trials to assess the impact of interventions for hypertension, preeclampsia, and primary kidney disease in pregnant women.

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BOARD REVIEW QUESTIONS

1. Which one of the following statements is correct regarding the normal physiologic changes of pregnancy?
   a. Glomerular filtration rate decreases
   b. Serum sodium increases to 4–5 mEq/L above non-pregnancy levels
   c. The plasma uric acid concentration increases
   d. There is a mild respiratory alkalosis, with a compensatory increase in renal excretion of bicarbonate

   Answer: d

   Rationale: Pregnancy is characterized by an increase in minute ventilation and mild respiratory alkalosis, with a fall in the partial pressure of carbon dioxide to approximately 27–32 mm Hg. There is an appropriate renal response, with increased renal excretion of bicarbonate. The glomerular filtration rate increases by approximately 40% in pregnancy. This leads to a decrease in uric acid concentration in normal pregnancy. Serum sodium falls, leading to mild hyponatremia, due to a decrease in the osmotic threshold for antidiuretic hormone release.

2. Which one of the following best describes the clinical syndrome of preeclampsia?
   a. Proteinuria is necessary to establish a diagnosis of preeclampsia
   b. Renal biopsy in preeclampsia is characterized by immune complexes within the glomeruli
   c. Glomerular endotheliosis, a hallmark of preeclampsia, occurs due to impairment of vascular endothelial growth factor (VEGF) signaling
   d. Patients with a history of preeclampsia do not have increased incidence of cardiovascular diseases in the long term

   Answer: c

   Rationale: The syndrome of preeclampsia is characterized by high circulating levels of soluble VEGF receptor [soluble fms-like tyrosine kinase-1 (sFlt1)] that induces glomerular capillary damage and endothelial swelling (glomerular endotheliosis) by inhibiting podocyte-derived VEGF signaling in the glomerulus.

3. A 29-year-old woman with a history of lupus nephritis presents to your office for follow-up. She received induction therapy for lupus nephritis approximately 1 year ago with CellCept, and is now in remission. She was recently found to be pregnant, and is currently 7 weeks' gestation. Her medications include CellCept 500 mg twice daily, plaquenil 200 mg daily, nifedipine 60 mg daily, aspirin 81 mg daily, and prednisone 5 mg daily. Her blood pressure is 140/85 mm Hg. Physical examination is unremarkable. Laboratory studies show serum creatinine 1.2 mg/dL (stable from prior measurements), and random urine protein-to-creatinine ratio 143 mg/g.

   Which of the following is the MOST appropriate next step in her management?
   a. Discontinue nifedipine and start methyldopa
   b. Decrease nifedipine to 30 mg daily
   c. Discontinue plaquenil and start azathioprine
   d. Discontinue CellCept and start azathioprine
   e. Discontinue aspirin

   Answer: d

   Rationale: Mycophenolate mofetil and mycophenolic acid, which are commonly used for the treatment of glomerular disease and kidney transplantation, are teratogenic and contraindicated in pregnancy. These agents should be discontinued prior to conception when pregnancy is anticipated; women of childbearing age who receive these medications should be counseled to use effective birth control. Women who conceive pregnancy while taking these agents should be switched immediately to agents compatible with pregnancy. For the maintenance of remission in lupus nephritis, azathioprine is the agent of choice. Prednisone is also safe and effective for maintenance or treatment of lupus nephritis in pregnancy. Discontinuation of plaquenil during pregnancy is associated with flare of extrarenal lupus, hence this agent should be continued. This patient's blood pressure is at target on her current dose of nifedipine, and this agent is safe and effective in pregnancy, so no changes in her antihypertensive regimen are indicated at this time. Low-dose aspirin reduces the risk of preeclampsia, and should be initiated (or continued) in women at increased risk for preeclampsia, such as this patient.

4. A 36-year-old woman with chronic hypertension presents to your office after a positive pregnancy test. Ultrasound confirms 10-week singleton gestation. Prior to pregnancy she was taking hydrochlorothiazide, which she stopped when she noted her positive pregnancy test. Her blood pressure in the office is 155/95 mm Hg.

   Which of the following would be the most appropriate agent to treat this patient's hypertension?
   a. Labetalol
   b. Captopril
   c. Hydrochlorothiazide
   d. Atenolol

   Answer: a.

   Rationale: Labetalol is a safe and appropriate antihypertensive agent for use in pregnancy. Methyldopa and dihydropyridine calcium channel blockers, such as long-acting nifedipine, are also appropriate for use in pregnancy. Angiotensin-converting enzyme inhibitors and angiotensin receptor antagonists are contraindicated in pregnancy, as they cause birth defects with second- and third-trimester exposure. Hydrochlorothiazide and other loop and thiazide diuretics are not teratogenic, but can theoretically interfere with the normal plasma volume expansion of pregnancy, and are generally avoided. Atenolol has been associated with impaired fetal growth in some studies; agents with combined α- and β-blockade, such as labetalol, maintain placental perfusion and are not associated with this adverse effect.

5. A 28-year-old woman presents at 32 weeks' gestation with a new onset of edema in her hands and face. Her blood pressure, previously normal, is 168/105 mm Hg. Blood pressure repeated 6 hours later is 150/98 mm Hg. Physical examination is notable for edema in her upper and lower extremities bilaterally. Which of the following findings would establish the diagnosis of preeclampsia?

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a. Uric acid > 8 mg/dL
b. Urine protein excretion > 150 mg on a 24-hour urine collection
c. Total bilirubin > 2.0 mg/dL
d. Platelet count < $100/mm^3$
e. The development of pedal edema

Answer: d

Rationale: In 2013, the American College of Obstetricians and Gynecologists (ACOG) published new diagnostic criteria for the diagnosis of preeclampsia. These criteria allow for the diagnosis of preeclampsia in women with the new onset of hypertension after 20 weeks' gestation, together with either proteinuria (>300 mg/day on 24-hour collection, or protein-to-creatinine ratio > 0.3 mg/mg on a random urine sample) or another sign or symptom of severe preeclampsia. These severe features include thrombocytopenia, elevated liver enzymes, acute kidney injury, cerebral, or visual symptoms. Hyperuricemia is a common feature of preeclampsia, but is not reliable for diagnosis and is not part of the diagnostic criteria. Transaminitis is a diagnostic feature of preeclampsia, but hyperbilirubinemia is not. Although pedal edema can be a symptom of preeclampsia, it is not considered a diagnostic criteria for preeclampsia according to the ACOG criteria.

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